

The institutions that absorb a shock

Three rigorous economics papers · two current LMIC studies and one Malawi field-experiment classic. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

The common thread is the institution households and citizens can still use when a larger shock arrives. In rural India, heat reduces farm resources and contracts women's marriage-migration options, especially where dowries are prevalent and credit is scarce. In Mali, an anticipated foreign intervention changes the expected return to democracy, taxation, and state capacity before the intervention itself occurs. In Malawi, locally governed savings groups turn small deposits into commitment, insurance, credit, and investment. **Literature trend:** recent applied work is moving beyond average shock effects toward the financial and political margins that transmit them. Panel comparisons, announcement timing, and cluster randomization differ in strength, but all three papers make institutions measurable mechanisms rather than vague context.

01 CURRENT / CLIMATE, GENDER, AND MIGRATION · STARTS ON PAGE 2 · World Bank Economic Review · online first 2026 · peer reviewed · CC BY 4.0

Scorching Heat and Shrinking Horizons: The Impact of Rising Temperatures on Marriages and Migration in Rural India

Does climate stress narrow women's geographic options at marriage? Using Indian censuses from 1991–2011, ERA5 temperature data, district fixed effects, state-by-year trends, and Conley standard errors, the authors estimate that a 0.1°C increase in long-run temperature reduces female rural-to-urban migration by 2.7 percent and rural-to-rural migration by 1.4 percent. Lower farm yields and consumption appear to tighten dowry and moving constraints, especially where dowries are common and formal credit is scarce. Three census waves and observational climate variation limit causal precision, but the paper makes a sharp case that warming can contract women's marriage markets through household balance sheets.

02 CURRENT / CONFLICT, INSTITUTIONS, AND PUBLIC FINANCE · STARTS ON PAGE 19 · Journal of Development Economics · online 2026 / forthcoming 2027 · peer reviewed · CC BY 4.0

Expected Foreign Military Intervention and Demand for State-Building: Evidence from Mali

Can an external security guarantee change citizens' demand for a capable state? The paper exploits Afrobarometer interviews conducted around the UN Security Council's December 2012 authorization of intervention in Mali, adding region-by-ethnicity fixed effects, event-time tests, weather and seasonality checks, distance-based IV estimates, and cross-country comparisons. After the announcement, expected democracy rose by 8 percentage points, tax morale by 0.26 standard deviations, and stated willingness to pay taxes by about 12 points. These are short-run reported attitudes rather than observed tax payments, so persistence is uncertain; still, the design shows how security expectations can shift the perceived return to state-building.

03 CLASSIC / DEVELOPMENT FINANCE AND HOUSEHOLD WELFARE · STARTS ON PAGE 38 · Journal of Development Economics 120 · 2016 · peer reviewed classic · CC BY 4.0

Impact of Village Savings and Loan Associations: Evidence from a Cluster Randomized Trial

Can a self-governed savings group substitute for missing rural finance? In 46 villages and roughly 1,775 households in northern Malawi, the authors randomized the timing of Village Savings and Loan Association training. Two years later, treatment villages showed higher savings and borrowing, more agricultural investment, improved food security, and greater income from existing small businesses. The intention-to-treat design is credible, although take-up reached only 45 percent and control-group participation rose to 21 percent, attenuating estimates. The classic is useful precisely because it shows how governance, commitment saving, and local intermediation can matter even when average business effects are uneven.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

Scorching heat and shrinking horizons

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Scorching Heat and Shrinking Horizons: The Impact of Rising Temperatures on Marriages and Migration in Rural India

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Abstract

This study delves into the gendered impacts of rising temperatures from climate change on long-term female migration patterns in rural India. We utilize a panel fixed-effects model to investigate the relationship between temperature levels and migration trends by employing a district-level panel dataset that combines Census data for years 1991, 2001, and 2011, multiple rounds of household surveys, and local weather measurements. Our results showcase statistically significant declining effects of rising temperatures on female rural-rural and rural-urban migration in the country. Specifically, the findings suggest that a 1 °C temperature increase is associated with a 22% decline in rural-urban and a 13% decline in rural-rural female migration in an average district in India. This decline is primarily due to diminishing marriage-related migration among women in the northern districts of the country, which have a historically high prevalence of dowry. We identify decreasing agricultural yields from rising temperatures as an underlying mechanism that is reducing resources to finance dowry in northern India. We further note that the declines in female migration are driven by districts with poor access to credit, and more so in the northern districts.

JEL Classification: J12; O15; Q54

Keywords: Climate change; Migration; Marriages; Dowry; India

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I Introduction

Anthropogenic climate change is one of the most pressing issues of our time, with far-reaching impacts on human society. According to the sixth assessment report of the Intergovernmental Panel on Climate Change (IPCC), global warming of 1.1 degrees Celsius above pre-industrial levels is already profoundly affecting human systems (IPCC, 2021). These effects are felt most acutely in underdeveloped countries located in low-latitude regions and near the equator (Castells-Quintana et al., 2017; Cline, 2007; Dell et al., 2012). In this regard, Carleton and Hsiang (2016) argue that climate change, by variably affecting different sub-populations like women and the poor, can lead to substantial changes in demographic structures in the long run. These effects can potentially result in further distortion of existing social and economic inequalities in developing countries.

One such demographic effect of climate change is on migration patterns, which has garnered significant attention in recent studies (Adger et al., 2015; Black et al., 2011; Cattaneo et al., 2019; Hoffmann et al., 2021). However, while the literature has extensively discussed the impacts of climate change on migration rates in the global south, there continues to be limited evidence on the gendered impacts on migration. This is critical, given the prevalence of various gender-related norms surrounding women's migration in developing countries in South Asia and Africa (Bau, 2021; Khalil & Mookerjee, 2018). In light of this, in this study, we investigate the long-run impacts of climate change on internal female migration in India and explore the mediating role of marriage in shaping those impacts. To do this, we create a unique database combining information on intra-district rural-urban and rural-rural female migration for the years 1991, 2001, and 2011 from the Census of India and district-level data on temperature and precipitation from ERA5 reanalysis weather data (Berrisford et al., 2011). Additionally, we gather multiple waves of household-level surveys by the National Sample Survey (NSS) of India between 1987 and 2011 to further delve into the dynamics.

The gendered effects of climate change on migration are quite significant. We find that a 1°C increase in mean decadal temperature is associated with a 22% decline in female rural-urban migration and a 13% decline in female rural-rural migration in an average Indian district. Our results are also robust to multiple sensitivity tests. They remain robust to the inclusion of state-year trends and when the spatial correlation between the districts is taken into account using Conley standard errors (Conley, 1999). The findings are critical as India has a patrilocal structure where most women migrate to their in-laws' houses after marriage (Chatterjee & Desai, 2020). In fact, most migrants in India are women who migrate because of marriage and not for economic reasons, which is also due to the low participation of women in labor markets (Afridi et al., 2018; Mehrotra & Parida, 2017).

Moreover, the event of marriage in India includes the traditional custom of dowry, in which the bride's household transfers wealth and resources to the groom's household during marriage. Despite legal procedures, the custom continues to be highly pervasive in the country (Chiplunkar & Weaver, 2023) and plays a decisive role in a woman's marriage. In this study, we showcase that the declining effects on female migration are driven by reductions in female marriage migration, primarily in the northern part of the country where the dowry custom has been historically more rooted (Mitchell & Soni, 2021). We find that falling agricultural yields due to rising temperatures (Taraz, 2018) that are eroding agricultural incomes act as an underlying mechanism driving our results. Moreover, we also document that poor access to credit in rural areas is moderating

the impacts of increasing temperatures on female migration. Besides, although we can only examine impacts on intra-district female migration due to data limitations, multiple sets of evidence indicate that most marriages occurred within the short distance of a woman's native place and within the same district during the time period of our study (Chiplunkar & Weaver, 2023; Fulford, 2013)¹, which upholds the findings of our study. The results of our study highlight a concerning trend as for many women in India, moving to the city through marriage is a means to broaden their horizons and improve their living standards (Kalpagam, 2008). Climate change, by interacting with traditional gender norm, is insidiously shrinking the prospects of women's movement to urban areas through marriage, trapping women in rural regions with potential negative consequences on their welfare.

Our study makes a significant contribution to multiple strands of literature. We supplement the literature on the demographic effects of climate change (Carleton & Hsiang, 2016). We provide evidence of how climate change is distorting demographic structure in the long run by disproportionately affecting the female sub-population in rural India. We also advance the growing literature on the effects of climate change on migration patterns in developing countries (Cattaneo et al., 2019; Hoffmann et al., 2021). Specifically, our study is related to works by Gray and Mueller (2012) and (Findley, 1994). Gray and Mueller (2012) examine the consequences of droughts on migration in rural Ethiopia, and note that drought events are associated with declines in female out-migration as households have fewer resources to bear wedding expenses as traditional norms dictate that the bride's household are responsible for spending on wedding expenses. Findley (1994) documents similar observations for Mali, where the occurrence of droughts is related to decreases in migration by women as resources to arrange weddings shrink. In our study, we show that rising temperature levels are associated with falling female marriage-related marriage migration in rural India. In this way, we also add to the literature on the broader economic implications of the increasing temperature levels in rural India. These studies include shrinking rural consumption levels (Aggarwal, 2020; Liu et al., 2023; Sedova et al., 2020), rising rural mortality rates (Burgess et al., 2014), and intensification of agricultural activities as demand for non-agricultural products shrivels due to falling agricultural incomes (Liu et al., 2023).

Moreover, our study augments the literature on how traditional gender-related norms shape gender outcomes in developing countries. For instance, Ashraf et al. (2019) document that in Indonesia and Zambia, groups that practice the custom of bride price tend to have more educated daughters. Lowes and Nunn (2017) note no associations between larger bride prices and earlier marriages and high fertility but even assert that bride price is linked to better quality marriages and high self-reported happiness from the wife. In contrast, in India, studies have recorded perverse gender outcomes due to dowry. For example, Bhalotra et al. (2020) detect that higher dowry costs, reflected in the high cost of gold, result in increased son preference behavior amongst the parents. Sekhri and Storeygard (2014) observe that dry rainfall shocks are linked to increases in the incidence of domestic violence against women and dowry deaths in India². This is because dowry payments are used to smooth consumption in the event of dry shocks by the groom's households. Moreover, Calvi and Keskar (2021) conclude that women

¹This is discussed further with evidence in Section II.II.

²According to Sekhri and Storeygard (2014), in India, a dowry death is legally defined as the death of a woman that occurs within seven years of her marriage and is caused by any burns or other bodily injury that do not occur under normal circumstances.

who paid higher dowries in India are less likely to be poor than those who did not pay. They also possess high control over resources in the household and have better bargaining power within the household. Calvi and Keskar (2021) add that gender poverty gaps are much more pronounced in households with no or low dowry payments relative to those with high and moderate dowry payments.

In this regard, we also supplement the literature on the interaction of climate change, marriage markets, traditional gender norms, and gender-related economic outcomes. Our findings closely align with those of Corno et al. (2020), who show that drought events increase the risk of early marriages in regions in Sub-Saharan Africa, where the bride price custom is common, in which the groom's household pays the bride during the event of marriage. They further show that droughts decrease the odds of early marriages in India as resources to finance dowry diminish. In congruence with our results, the declines in their study are driven by regions in northern India, where dowry custom is ubiquitous. Another related study by Trinh and Zhang (2021) contrasts Vietnam, which has a tradition of bride price, and India and records similar outcomes that rainfall shocks soar the likelihood of child marriage in Vietnam but have an opposite effect in India.

The rest of this article is organized as follows. Section II discusses the study context, and III provides information on data sources and the methodology. Section IV discusses results, and Section V delineates the underlying mechanism and contextual factor shaping our results. Section VI presents the discussion and policy implications. Lastly, Section VII provides the concluding remarks.

II Study Context

In this section, we provide some background on the agricultural sector and climate change, female marriage migration, and the dowry custom in India.

II.I Agricultural Sector and Climate Change in India

Despite the falling contribution of the agricultural sector to the national Gross Domestic Product (GDP) of India from 30% in 1981 to 16.5% in 2019 (Gulati & Juneja, 2022), the sector remains a crucial component of the economy. It is a major employer of the rural population (Chand et al., 2017) and employed 43% of the country's workforce and 62% of the rural population in 2019. It is also a principal source of employment for women in rural areas, with 76% of women working in this sector as of 2011-12 (Gulati & Juneja, 2022).

Given the strong dependence on the agricultural sector, climate change poses a significant threat to the Indian economy. Studies show a consistent rise in average temperature levels across the country (Kumar et al., 2006; van Oldenborgh et al., 2018), with rainfall trends remaining relatively unchanged on a national scale but exhibiting more region-specific variations (Dash & Hunt, 2007; Kumar et al., 2006; Mall et al., 2006). Multiple studies have highlighted the negative impact of rising temperatures on agricultural yields in India. Taraz (2018) finds that high temperatures are damaging agricultural yields in all districts of the country, particularly the historically colder districts. This is because hotter districts have relatively better accessibility to inter- and intra-crop adaptation methods that aid in adapting to moderate increases in temperature levels. Another study discusses how temperature spikes of 3-6 °C in the Indo-Gangetic

III.I Data Sources

We use three main data sources in the study and assemble a district-level panel dataset by combining data on migration rates and weather variables. First, we formulate district-level measures of female rural-rural and rural-urban migration rates using the Indian Census data. Second, we aggregate the gridded monthly weather data published by the European Centre for Medium Range Weather Forecasts (ECMWF) (Berrisford et al., 2011) to construct district-level measures of temperature and precipitation. Third, we utilize micro-level household survey data from the National Sample Surveys (NSS) of India, conducted between 1987 and 2012. Using NSS, we extract district-level information on marriage migration by women. Together, these data sources allow us to construct a rich dataset to examine the impact of rising temperatures on female migration patterns in rural India and to test the effects on marriage-related migration.

III.I.1 Data on climate

For the climate data, we employ gridded monthly data on temperature and precipitation from the ERA5 high-quality reanalysis data produced by the European Centre for Medium Range Weather Forecasts (ECMWF) (Berrisford et al., 2011). Following the recent climate literature, we compute the average value of all grid points falling within a district boundary. Besides, as a robustness test in the later sections (Section IV.I), we also use an alternative aggregation method in which the grid-level data is aggregated to district-level weather outcomes, that is, mean temperature and mean precipitation by calculating the weighted average of all grid points within 100 kilometers of each district's geographic center using inverse-square weighting (Taraz, 2018).

The main variables are annual average temperature and average precipitation at the district level, and these variables are computed for each year between 1981 and 2010. The spatial distribution of the changes in the temperature and precipitation between the years 1981 and 2010 is presented in Figure 1. It is clear that almost all districts have experienced a rise in average temperature levels by at least 0.1 °C during the period of analysis. However, the figure shows a substantial spatial difference in temperature increases that can be utilized for our study. The descriptive statistics of these variables are presented in Table A1 in Appendix A.

III.I.2 Data on female rural-rural and rural-urban migration

To construct the main dependent variables capturing the female rural out-migration rates, we use the Census of India data for the years 1991, 2001, and 2011. We gather the district-wise population data using the Primary Census Abstract (PCA) data tables. The data on migration are compiled from the Migration (D-series) data tables from the Census digital library⁵. These tables provide district-level data on the total female population and female intra-district rural-urban and rural-rural migrants. For completeness and comparison, we have also collected information on the total and male population in the district and total and male intra-district rural-urban and rural-rural migrants.

To ensure consistency in district boundaries across the years, we harmonize the district boundaries using the district mapping file provided by Kumar and Somanathan (2017). There were 466 districts in India in 1991. Figure A.1 in the Appendix depicts these 466 district boundaries that are consistent between 1991, 2001, and 2011. We could not include 14 districts from the state of Jammu and Kashmir⁶ in the analysis due to the unavailability of census data in 1991. Additionally, due to the unavailability of weather data, we dropped 4 districts of union territory Pudducherry and 1 district of Lakshadweep, leading to a final sample of 447 districts.

The primary dependent variable is defined as the share of rural-urban (rural-rural) female intra-district migrants in the total female population of a district. Specifically, it is defined as:

$$Y_{j,t} = \frac{M_{j,t}}{P_{j,t}} \quad (1)$$

where $M_{j,t}$ is the number of rural-urban (rural-rural) female migrants and $P_{j,t}$ is the total female population in district j and in the decade beginning with year $t (= 1991, 2001, 2011)$. We construct this variable for female intra-district rural-urban (rural-rural) migrants as it is our primary outcome of interest. For comparison, we also construct similar variables measuring involving total and male intra-district rural-urban and rural-rural migrants.

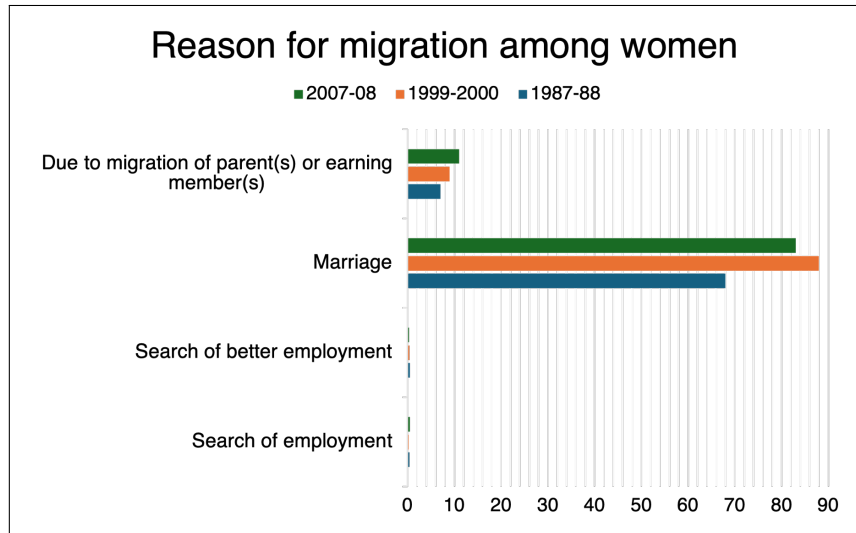
We present these variables in Figure 2, which highlights two important aspects of internal migration in India: (1) Intra-district rural-rural migration by both men and women is more common than rural-urban migration; and, as previously discussed (2) Women make up the majority of internal migrants from rural areas in the country, irrespective of whether they move to other rural or urban destinations.

⁵<https://censusindia.gov.in>

⁶The state now is organized into two union territories: Jammu and Kashmir and Ladakh

NSS-EUS provides additional evidence that most internal migration by working-age⁸ married women are due to marriage. As shown in Figure 3, 68% of internal married working-age female migrants in India moved due to marriage in the 1987-88 round, which increased to over 80% in subsequent rounds. Additional descriptive statistics derived from NSS data are displayed in Figure A.3 in Appendix A.

Figure 3: Female Migration by Reason in India, NSS, 1987-2008



Note: The chart depicts the primary reason for migration among married working-age women in three NSS rounds: 1987-88, 1999-2000, and 2007-08. Source: Authors' own illustration using NSS-EUS data.

We establish a linkage between the NSS and Census data by utilizing the district codes available in the NSS surveys. To ensure that the districts are consistent with the Census data and to maintain the same district boundaries, we refer to Murthi et al. (2001) and Kumar and Somanathan (2017) .

III.I.3 Data on additional variables

The analysis is supplemented with data from several other sources. As we argue that reductions in agricultural yields due to rising temperatures act as an underlying mechanism explaining our results, we gather data on agricultural yields to test this mechanism. We collect agricultural data from the Village Dynamics in South Asia (VDSA), provided by the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT, 2015). This data set has annual measures of district-level agricultural yields for 15 crops between the years 1966 and 2010. These crops are rice, wheat, sugarcane, cotton, groundnut, sorghum, maize, pearl millet, finger millet, barley, chickpeas, pigeon pea, sesame, rapeseed and mustard, castor, and linseed. Following Taraz (2018) , we derive a price-weighted composite yield measure at the district level for each year

of women has fallen in the same time frame. Rao and Finnoff (2015) have conclusively shown that these trends are indeed due to rises in marriage migration and not increases in migration by women for economic purposes that are disguised as marriage migration. Mazumdar et al. (2013) also discuss this issue and using primary survey data and field studies conclude that labor migration by women in India is very low and most migration by women is for marriage.

⁸Following Rao and Finnoff (2015) , we consider working-age to be between 15 and 64 years old.

climate change literature on India, which documents consistent rises in average temperatures across the country but region-specific changes in precipitation, we only interpret the coefficients associated with the temperature levels throughout this study. Thus, β_0 is the main coefficient of interest in this study and captures the effect of rising temperatures on migration rates, and $\beta_0 < 0$ signifies that migration rates are declining with rising temperatures.

Furthermore, we also include fixed effects, ϕ_j , capturing fixed district characteristics. γ_t is the year effects that control for changes over time that are consistent across districts. We only include fixed effects as controls to ensure that the model stays parsimonious and to avoid overcontrolling. This is because including other controls, such as socioeconomic characteristics that might be correlated to agricultural productivity, may introduce bias in the estimation due to over-controlling (Cattaneo & Peri, 2016). The standard errors are clustered at the district level and are provided in parentheses in the regression tables.

Specification (2) is a reduced-form linear relationship between rural-urban and rural-rural female migration rates and climate variables. In the upcoming sections, we discuss the possible channels through which climate change may shape migration rates, such as changes in agricultural productivity and contextual factors like access to credit.

IV Results

Table 1 illustrates the estimated effects of annual temperatures on intra-district female rural-rural and rural-urban migration rates in India. They indicate a statistically significant negative effect of increasing temperatures on rural-rural and rural-urban female migration rates. Specifically, the results show that a 1 °Celsius increase in temperature in an average Indian district is associated with a 22% decrease in the share of female rural-urban migrants and a 13% reduction in the share of female rural-rural migrants. The effects on male migration rates are not statistically significant, aligning with the findings of Liu et al. (2023) .

Table 1: Effect of Rising Temperatures on Migration, 1991-2011

	Rural-Urban			Rural-Rural		
	Total	Male	Female	Total	Male	Female
	(1)	(2)	(3)	(4)	(5)	(6)
Temperature	−0.004 (0.003)	−0.002 (0.003)	−0.007 (0.003)**	−0.018 (0.008)**	−0.006 (0.006)	−0.030 (0.012)**
Precipitation	0.004 (0.003)	0.003 (0.002)	0.004 (0.003)	−0.001 (0.005)	0.008 (0.003)**	−0.010 (0.007)
District FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Observations	1,341	1,341	1,341	1,341	1,341	1,341

Note: The dependent variable in (1) is the share of total intra-district rural-urban migrants, in (2) is the share of male intra-district rural-urban migrants, in (3) is the share of female intra-district rural-urban migrants, in (4) is the share of total intra-district rural-rural migrants, in (5) is the share of male intra-district rural-rural migrants, and in (6) is the share of female intra-district rural-rural migrations, in years 1991, 2001, and 2011. The independent variables are the decadal averages of annual temperature (°C) and annual precipitation (mm). The standard errors are clustered by district in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

We argue that the declines in female migration are primarily due to the decreases in marriage-related migration. Furthermore, these declines could be driven by the districts in northern India, where the custom of dowry, as previously discussed, has been historically widespread. To check for this, we categorize the districts as northern districts that are located in the northern states¹¹. We exploit this heterogeneity in the historical prevalence of dowry between the northern districts and the rest of India by re-running the previous analysis and allowing the coefficients of temperature and precipitation to vary depending on whether the district is in a northern state. The results are reported in Table 2 and match our expectations. In the northern districts, rising temperatures have a pronounced negative effect on female migration rates¹².

Table 2: Effect of Rising Temperatures on Female Migration in Northern Districts, 1991-2011

	Rural-Urban	Rural-Rural
	Female	Female
	(1)	(2)
Temperature	0.010 (0.005)**	0.029 (0.016)*
Temperature x North	-0.022 (0.004)***	-0.075 (0.012)***
Precipitation	0.006 (0.003)*	-0.007 (0.006)
Precipitation x North	-0.017 (0.004)***	-0.040 (0.015)***
District FE	Y	Y
Year FE	Y	Y
Observations	1,341	1,341

Note: The dependent variable in (1) is the share of female intra-district rural-urban migrants in the years 1991, 2001, and 2011. The dependent variable in (2) is the share of female intra-district rural-rural migrants in the years 1991, 2001, and 2011. The independent variables in (1) and (2) are the decadal averages of annual temperature (°C) and decadal averages of annual temperature interacted with “North” dummy denoting 1 if the district lies in a major northern state such as Punjab, Uttarakhand, Uttar Pradesh, Haryana, Bihar, Jharkhand, Rajasthan, Madhya Pradesh, and Chhattisgarh, and 0 otherwise. It also includes the decadal averages of annual precipitation (in mm) and decadal averages of annual precipitation interacted with the “North” dummy. The standard errors are clustered by district in parentheses.

*p < 0.10, **p < 0.05, ***p < 0.01

It might still be the case that these changes observed through Census data are not due to

¹¹We follow the Government of India’s categorization of the north and north-central cultural zone (Ministry of Culture, Govt. of India, [n.d.](#)) to identify the northern states and classify the states Punjab, Haryana, Uttar Pradesh, Rajasthan, Bihar, Rajasthan, Uttarakhand, Chhattisgarh, Jharkhand, and Madhya Pradesh as northern states. We exclude the erstwhile state of Jammu & Kashmir due to the unavailability of Census data in 1991. The state of Himachal Pradesh is also omitted as it has a smaller population and area, and we only consider the major northern states. The state of Uttarakhand is also small but has been considered as it was part of the state of Uttar Pradesh until 2000. Similarly, the states Jharkhand and Chhattisgarh are also considered as they belonged to the states of Bihar and Madhya Pradesh, respectively, until the year 2000.

¹²In an additional analysis, we allow the temperature coefficients to vary based on whether the district is in southern states of India, where the custom historically has not been prevalent. The results are reported in Appendix C, highlighting no negative effects for southern districts. These results complement the findings of Corno et al. (2020) .

changes in marriage-related female migration. Therefore, we further corroborate our analysis by providing more evidence on the marriage channel using the NSS-EUS survey for the years 1987-88, 1999-2000, and 2006-07. These waves of EUS had modules on migration where the survey respondents, that is, the members of the household, were asked if their current place of enumeration was different from their last usual place of residence. If so, the primary reason for this migration is recorded. Thus, using this data and the survey weights provided by NSS, for each of these waves of EUS, we construct an outcome variable that is the share of the total married working-age female migrants in the district who have undergone intra-district rural-urban migration due to marriage in last ten years. We compute similar measures for intra-district rural-rural migration. The approach is aligned with the measurement of migration variables using the Census data, but here, the variables are refined further to capture only the marriage-related migration by women.

The computation of climate variables stays the same. We then carry out a panel fixed effects regression in a similar manner as before with district-fixed and year effects. The results are presented in Table 3, and findings from regression (3) indicate a negative association between increases in temperature and female rural-urban marriage migration in the northern regions. These findings lend further support to the initial results, suggesting that the declines in female migration observed previously are driven by decreases in marriage-related migration in northern India.

Table 3: Effects of Rising Temperatures on Marriage Migration, NSS Survey, 1987-2007

	Rural-Urban	Rural-Rural	Rural-Urban	Rural-Rural
	Female	Female	Female	Female
	(1)	(2)	(3)	(4)
Temperature	0.009 (0.016)	0.0005 (0.044)	0.030 (0.017)*	-0.001 (0.053)
Temperature x North			-0.049 (0.023)**	-0.007 (0.069)
Precipitation	0.001 (0.015)	-0.023 (0.033)	0.022 (0.015)	-0.013 (0.034)
Precipitation x North			-0.026 (0.023)	-0.070 (0.059)
District FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Observations	881	881	881	881

Note: The dependent variables in (1) and (3) are the share of married female intra-district rural-urban migrants of working age who have migrated due to marriage in NSS rounds 1987-88, 1999-00, and 2007-08, respectively. The dependent variables in (2) and (4) are the share of married female intra-district rural-rural migrants of working age who have migrated due to marriage in NSS rounds 1987-88, 1999-00, and 2007-08, respectively. The independent variables in (1) and (2) are the decadal averages of annual temperature ($^{\circ}\text{C}$) and precipitation (in mm). The independent variables in (3) and (4) are the decadal averages of annual temperature ($^{\circ}\text{C}$) and decadal averages of annual temperature interacted with “North” dummy denoting 1 if the district lies in a major northern state such as Punjab, Uttarakhand, Uttar Pradesh, Haryana, Bihar, Jharkhand, Rajasthan, Madhya Pradesh, and Chhattisgarh, and 0 otherwise. The regression also controls for the decadal average of precipitation (in mm) and decadal average of precipitation interacted with “North” dummy. The standard errors are clustered by district and are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Moreover, we posit that these declines in marriage migration by women are due to the

diminishing agricultural income and wealth from soaring temperatures in rural areas, which, in turn, is dwindling the resources for dowry in northern areas. We provide suggestive evidence of shrinking dowry amounts in rural northern India in figure 4 below. The figure displays a negative relationship between net dowry paid by brides' households and rising temperatures.

IV.I Robustness Tests

In this section, we discuss the robustness tests we implement to further validate our main results regarding female migration rates in Table 1. These tests include adding state-year trends to the regression equation, re-running the analysis using alternative climate variables, and accounting for spatial autocorrelation. The results of robustness tests are attached in Appendix B.

State-year trends. In the first robustness test, we add state-year trends to our regression equation to capture state-specific time trends. The findings, presented in Table A3, show a slight increase in the magnitude of the estimates. Importantly, the direction and statistical significance of the results stay consistent with the initial findings.

Alternative climate variables. We explore the sensitivity of our main results to different measures of temperature and precipitation. For this, we aggregate grid-level weather data to district-level outcomes by calculating the weighted average of all grid points within 100 kilometers of each district's geographic center, using an inverse-square weighting of the distances, as suggested by Taraz (2018). These results, which are displayed in Table A4, affirm the robustness of the initial findings. The estimates remain statistically significant and consistent with the main results. The magnitudes of the estimated coefficients have increased slightly, indicating that the effects of rising temperature on female migration rates might be even higher than previously estimated.

Conley standard errors. In the third robustness test, we adjust the standard errors to consider possible spatial auto-correlation (Conley, 1999; Hsiang, 2010). We follow Sekhri and Storeygard (2014) and set the distance cutoffs at 100, 150, and 200 kilometers. The results, detailed in Table A5, demonstrate that the estimates remain robust when applying Conley standard error adjustments for these distance cutoffs.

V Mechanisms and contextual factors

In this section, we first investigate the underlying mechanism driving our results. We explain that declines in agricultural yields due to rising temperatures are eroding the wealth of rural households. This, in turn, harms their ability to finance the dowry payments. In addition, we test whether the effects of rising temperatures on female migration rates vary with a contextual factor, such as poor access to credit.

V.I Agricultural Yields and Dowries

As we discussed before, evidence supported by simulation models and observational data shows the detrimental effects of temperature rises on agricultural yields in India. We replicate these findings for the time period of our study and, specifically, carry out an analysis to dissect the effects for the northern districts. We follow Taraz (2018) and Liu et al. (2023) to measure the yearly aggregate agricultural yield at the district level. As detailed in Section III, we use the VDSA data on agricultural yields and compute the aggregate agricultural yield at the district

level for the time period of our analysis, 1981-2010. Thereafter, we conduct a panel fixed-effects estimation with the natural logarithm of the aggregate agricultural yields as the dependent variable¹³.

As expected, the results in Table 4 show a marked decline in agricultural yields due to rising temperatures.

Table 4: Effects of Rising Temperatures on Agricultural Yields, 1981-2010

	Log Agr. Yield
	(1)
Temperature	−0.065 (0.019) ^{***}
Precipitation	0.044 (0.010) ^{***}
District FE	Y
Year FE	Y
Observations	8,693

Note: The dependent variables in (1) is the natural logarithm of the aggregate agricultural yields, between the years 1981 and 2010. The independent variables in (1) are the average annual temperature (°C) and precipitation (in mm). The standard errors are clustered by district and are in parentheses.

* p < 0.10, ** p < 0.05, *** p < 0.01

How do these declines in agricultural yields translate into lower resources for dowry? The gradual declines in agricultural yields from rising temperatures over the years are decreasing income for agricultural households. However, rural households in India exhibit consumption patterns characterized by habit formation, meaning they slowly adjust their consumption in response to income volatility (Khanal et al., 2019), often delving into their savings to finance consumption. As the agricultural households gradually and disproportionately adjust their consumption to the decreases in agricultural incomes, far fewer resources are left for savings for dowry¹⁴.

The social and economic costs of not being able to get a daughter married are high (Anukriti et al., 2022), which can compel a few agricultural households to possibly overcome these losses in savings by adapting to rising temperatures and finding ways to save enough for dowry. However, this could be specifically more difficult for relatively poorer households¹⁵, who are unable to adapt and fail to save enough. As the cost of an unmarried daughter is high, these households

¹³We execute the following econometric analysis:

$$\ln Y_{j,t} = \eta_0 T_{j,t} + \eta_1 P_{j,t} + \phi_j + \gamma_t + \epsilon_{j,t}$$

Here, $Y_{j,t}$ is the aggregate agricultural yield in district j and year t . $T_{j,t}$ is the annual average temperature in °C and $P_{j,t}$ is the annual average precipitation in mm in district j and year t . We control for district fixed effects using ϕ_j and year effects through γ_t .

¹⁴We cannot completely discount that consumption levels are not changing. Studies have discussed the negative effects of rising temperatures on the consumption of rural households (Liu et al., 2023). More likely, both consumption and savings are drying up.

¹⁵Approximately 70% of the rural population in India has zero to little landholdings (Asher et al., 2022). Research shows considerable linkages between the size of landholdings and access to adaptation to climate change (Baland & Robinson, 2008; Bryan et al., 2009; Deressa et al., 2009; Dustmann & Okatenko, 2014).

save in advance and enable borrowing during the marriage year to finance dowry and wedding expenses (Anukriti et al., 2022).

For this analysis, we identify districts with low access to banks in the baseline year 1980. We determine whether the per capita number of bank branches was below the national median and create a binary variable to represent this condition: it equals 1 when the per capita number of bank branches in a district in 1980 is below the national median and 0 otherwise. The spatial plot in Figure A.4 in the Appendix shows significant variation in access to bank branches across districts in 1980. To assess how it moderates the effect of rising temperatures on female migration rates, we carry out a variation of the regression equation (2). We allow the coefficients attached to temperature to change based on whether the district had poor access to credit at the baseline.

The findings in Table 5 outline that limited access to banks significantly shapes the impact of temperature increases on female migration, both rural-rural and rural-urban, with a more substantial effect on the latter. Our results confirm that the access to credit constraints faced by poor agricultural households prevent them from saving for dowry in advance, challenge securing loans during the marriage years, and perhaps even constrain their ability to adapt to rising temperatures. This, in turn, is intensifying the effect of climate change on the marriage prospects of rural women and the marriage-related migration by them. In magnitude terms, these results show that a 1 °Celsius increase in temperature is associated with a 25% decline in female rural-urban migration in districts with low access to banks and a 19% decrease in female rural-rural migration. The lower magnitude of the declining effect associated with rural-rural female migration hints that rural households in districts with poor access to credit are perhaps resorting to marrying their daughters to other rural areas rather than to urban areas by offering lower dowry amounts.

Additionally, we adjust the analysis to examine the effects specifically in the northern districts. The findings are depicted as results of regressions (3) and (4) in Table 5. Unsurprisingly, we observe that the diminishing effects are driven by northern districts and, more so, by northern districts with limited access to credit at the baseline.

women’s welfare (Calvi & Keskar, 2021, 2023). On a different note, Chiplunkar and Weaver (2023) ’s recent study suggests that, as India’s literacy rate is rising sharply, the dowry custom might contract in the future as women’s literacy improves and the supply of educated grooms increases. However, other studies present a contrasting view, arguing that dowry custom would continue to play a fundamental role in marriages in India in the coming years (Beauchamp et al., 2017). This is because grooms increasingly seek educated women as brides but not highly educated. Thus, highly educated women may face challenges in finding suitable grooms and might need to offer substantial dowries for marriage.

Nonetheless, an important policy recommendation based on the findings is improving access to credit in rural areas. Burgess et al. (2014) have demonstrated how access to banks can reduce mortality in rural areas during summers in India. Banks facilitate consumption smoothing and provide loans at manageable interest rates, which can be crucial for financing dowry resources and adapting to climate change. For instance, farmers can obtain credit to invest in adaptation strategies such as irrigation, crop diversification, mechanization, and purchasing seeds. Additionally, banking access also offers resources for migration to cities. This has been highlighted by Bryan et al. (2014) , who found that offering small loans to subsistence-based farmers and job matching can reduce their aversion to migration. However, it is critical that financial inclusion policies are gender-sensitive. This ensures that women have equal opportunities to migrate for education or work, in addition to traditional marriage-related migration. Mehrotra and Parida (2017) argue that women’s labor force participation is likely to increase as more women attain education. Yet, this progression depends on women’s mobility and sense of safety when moving out. Therefore, policy interventions are required to first encourage the acceptance of women migrating for reasons other than marriage, such as education or employment opportunities, and make destination places safer. Furthermore, as suggested by Anukriti et al. (2022) , policies that liquidate dowry, such as financial inclusion policies that mitigate barriers to savings and induce savings behavior in poor rural households, can evade distress borrowing at exorbitant rates from informal sources during the time of marriage.

Finally, our study highlights the urgent need for state interventions to transform rural economies to be more resilient to climate change. Infrastructure programs, like road construction and electrification that connect rural areas to cities, can enhance access to agricultural markets, reduce information costs, and bolster the uptake of adaptation means by agricultural households.

VII Conclusion

The literature on climate change and migration in developing countries has expanded significantly in recent years. However, there remains a notable gap in understanding the specific impacts on women, who often constitute a major share of internal migrants due to the patrilocal structure prevalent in these regions. Although female migration appears non-economic, there are often various economic channels underlying the marriage of a woman in these countries. In our study, we demonstrate that rising temperatures are decreasing female migration rates in India, specifically female marriage-related migration in northern regions. We contend that this is due to the historical prevalence of dowry in northern India and identify decreasing agricultural yields from soaring temperatures as an underlying mechanism. In particular, we show that the reductions in agricultural yields are shrinking resources to finance dowry. We further note the

declining effects are much more pronounced in regions with limited access to credit.

It should be noted that various norms surrounding marriages, other than dowry, in the country remained largely unchanged during the time period of our analysis, which upholds the validity of our results. For instance, marriages continue to be arranged by parents with little say from the daughters in rural areas, a trend that persisted despite societal modernization and contrasts with predictions of theories of family change (Allendorf & Pandian, 2016; Banerji & Deshpande, 2021). Additionally, marriages continue to occur within the same caste and religion (Allendorf & Pandian, 2016; Banerjee et al., 2013). Comparing 1984 and 2009, Kamble et al. (2014) show that despite massive cultural shifts, mate preferences in India have persisted, including the preference for grooms with higher perceived economic security by women. These factors further support the results of our study.

Nevertheless, our study has important limitations. Firstly, although we have provided ample evidence showing that most marriages occur within short distances of native places during the time period of our study and within districts, we cannot completely discount the possibility that climate change might be inducing rural households to marry off their daughters to far-off places with different weather as a risk diversification mechanism (Rosenzweig & Stark, 1989). This is an issue that could not be explored in our study due to the unavailability of data from the Census of India and NSS, but it is a useful avenue for future research. Secondly, it might be the case that village endogamy, or marrying within one's own village, became more common during our analysis period, which could not be captured through migration rates. In this context, we rely on evidence provided by Mazumdar et al. (2013), who, based on their primary survey between 2009 and 2011, highlight a contrasting trend of increasing exogamous marriages in previously endogamous regions and note an overall rise in exogamous marriages among younger cohorts.

Our study makes a crucial contribution to the literature on the gendered economic effects of climate change and underlines how climate change can potentially increase economic inequalities by disproportionately affecting women. The findings of our study can be utilized in future studies to investigate further the broader dynamics of climate change impacts on marriages, female migration, and gender outcomes in other developing countries where marriage-related payments such as dowry and bride price are widespread. Moreover, future research should also examine how such traditional gender norms in interaction with climate change are shaping other economic outcomes, such as educational attainment, and social outcomes, such as domestic violence, for women in developing countries.

Table 5: Effect of Rising Temperatures on Female Migration and Heterogeneous Effects by Access to Banks, 1991-2011

	Rural-Urban	Rural-Rural	Rural-Urban	Rural-Rural
	Female	Female	Female	Female
	(1)	(2)	(3)	(4)
T	0.005	-0.019	0.010	0.028
	(0.004)	(0.013)	(0.005)*	(0.016)*
T x Low Bank Access	-0.011	-0.037		
	(0.003)***	(0.012)***		
T x North			-0.018	-0.057
			(0.005)***	(0.013)***
T x North x Low Bank Access			-0.006	-0.034
			(0.002)***	(0.013)***
P	0.016	-0.016	0.006	-0.006
	(0.005)***	(0.007)**	(0.003)*	(0.006)
P x Low Bank Access	-0.020	0.008		
	(0.005)***	(0.011)		
P x North			-0.012	-0.021
			(0.004)***	(0.022)
P x North x Low Bank Access			-0.004	-0.010
			(0.004)	(0.028)
District FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Observations	1,326	1,326	1,326	1,326

Note: The dependent variables in (1) and (3) are the share female intra-district rural-urban migrants, respectively, in years 1991, 2001, and 2011. The dependent variables in (2) and (4) are the share female intra-district rural-rural migrants, respectively, in years 1991, 2001, and 2011. The independent variables in (1) and (2) are the decadal averages of annual temperature ($^{\circ}\text{C}$), decadal averages of annual temperature interacted with “Low Bank Access” dummy denoting 1 if the per capita Bank branches at the baseline was lower than the national median and 0 otherwise, decadal average of annual precipitation (in mm), and decadal average of annual precipitation (in mm) interacted with the “Low Bank Access” dummy. The independent variables in (3) and (4) are the decadal averages of annual temperature ($^{\circ}\text{C}$), decadal averages of annual temperature interacted with “North” dummy denoting 1 if the district lies in a major northern state such as Punjab, Uttarakhand, Uttar Pradesh, Haryana, Bihar, Jharkhand, Rajasthan, Madhya Pradesh, and Chattisgarh, and 0 otherwise, and decadal averages of annual temperature interacted with “North” dummy and “Low Bank Access” dummy. The regressions also control for the decadal average of annual precipitation (in mm), and decadal average of annual precipitation interacted with the “North” dummy, and decadal average of annual precipitation interacted with the “Low Bank Access” dummy. The standard errors are clustered by district in parentheses.

*p < 0.10, **p < 0.05, ***p < 0.01

VI Discussion & Policy Implications

In this study, we discuss the gender-specific impacts of climate change, particularly the effects of rising temperatures, on long-term female migration patterns in rural India. We provide evidence of temperature increases substantially affecting female rural-rural and rural-urban migration rates. These reductions in female migration rates are mainly driven by declines in marriage-related female migration in northern districts where the custom of dowry is ubiquitous.

The policy implications of this study, particularly concerning the traditional custom of dowry, are multifaceted and complex. Various studies have highlighted the ineffectiveness of the Dowry Prohibition Act of India, with some suggesting legal procedures can inadvertently exacerbate



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EXPECTED FOREIGN MILITARY
INTERVENTION AND DEMAND FOR
STATE-BUILDING: EVIDENCE FROM MALI

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Abstract

We study the informational effects of foreign military intervention on citizens' motivations to participate in state-building processes. We analyze the 2012 United Nations Security Council resolution that authorized intervention in Mali to reunify the country and restore democracy during a profound institutional crisis. By exploiting the randomness of the foreign intervention announcement, relative to the timeline of the Afrobarometer interviews, we document that individuals interviewed the days after the announcement have a higher intrinsic motivation to comply with taxes, are less inclined to refuse to filing taxes, and are more reluctant to evade even if had chances, relative to individuals, interviewed immediately before, with the same characteristics, region, and ethnic group. We demonstrate that these effects are specific to regions characterized by low state capacity and limited ethnic diversity, as well as to individuals who perceive that their ethnic group has not been systematically discriminated against by the state. Consistently with our story, we document that motivations to comply only increase in respondents with access to the news at home (who own either a TV or a radio). Our results survive a wide range of falsification tests and indicate that foreign military interventions signal state-building, raising the expected benefits to participate.

JEL Class.: H11, H26, H56, D74.

Keywords: Foreign Military Intervention, State Capacity, Tax Morale, Weak States, Mali.

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Expected Foreign Military Intervention and Demand for State-Building: Evidence from Mali *

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1 Introduction

There is an established consensus that political institutions are an essential determinant of economic growth ([Acemoglu, Johnson and Robinson, 2001](#); [Acemoglu and Robinson, 2012](#)). Specifically, it is the power of the state to raise taxes and provide public goods (in short, state capacity) that plays a key role in the process of development and in sustaining high income levels ([Besley and Persson, 2011](#)). A number of works have studied and provided evidence about the determinants of weak state capacity that can often be traced to the absence of common interests or the existence

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of strong political conflicts ([Besley and Persson, 2009, 2014a](#); [Acemoglu, Ticchi and Vindigni, 2011](#)). It is also understood that high levels of fiscal capacity come not only from an efficient enforcement system based on monitoring and punishment, but are also the result of a high intrinsic motivation of citizens to comply with taxes ([Besley and Persson, 2014b](#); [Besley, 2020](#)) and, more generally, of the citizens' attitude toward the state ([Callen, Weigel and Yuchtman, 2024](#)). What instead is less clear are the determinants of the citizens' intrinsic motivation to pay taxes, often called tax morale ([Luttmer and Singhal, 2014](#)),¹ and how such incentives interact with the material ones ([Besley, Jensen and Persson, 2023](#)); this is also because poor countries are generally characterized by both inefficient bureaucracies and low motivation to comply with taxes, while the opposite holds in rich economies. Therefore, weak states that want to invest in institution-building face an important compliance problem ([Besley and Dray, 2024](#)): on the one hand, citizens have low motivation to cooperate and to participate to the provision of public goods and services by paying more taxes while, on the other hand, with little voluntary compliance by citizens, the state's possibilities of collecting taxes will have to rely predominantly on coercion, a circumstance which is hardly feasible, especially for states with weak capabilities.²

This paper provides a contribution to these issues by focusing on the understanding of the factors affecting the intrinsic motivation to pay taxes and emphasizing the potential role played by shocks to citizens' attitude toward the state in the process of state-building. Specifically, our research shows that the expectation of a foreign military intervention aiming at helping the government to take back control of territories and rebuild a well-functioning state apparatus can be an important state-building signal for citizens who, in turn, react with an increase in their motivations to comply

¹[Luttmer and Singhal \(2014\)](#) provide a survey for relevant studies on tax morale that they define as an umbrella term capturing non-pecuniary motivations to comply with taxes. Luttmer and Singhal also note and illustrate many examples that tax authorities in developing countries are indeed much concerned about increasing voluntary compliance with tax laws.

²It is worth emphasizing that despite the critical role institutions accomplish in promoting economic development, citizens' demand for state-building is stagnant ([Robinson, 2023](#)). According to the available survey data, a close to majority of Sub-Saharan Africans prefer to pay low taxes, have low public services, and to pursue economic growth without investments in the tax system. According to the data provided by [Afrobarometer \(2015\)](#), around one-third of respondents in that area of the continent would prefer to attain economic growth not through an increase in fiscal capacity but through other (not specified) means. This share is larger than 40% in Botswana, Burundi, Guinea, Lesotho, Malawi, Namibia, Tanzania, Uganda, and Zimbabwe.

with tax duties. This result is important for two reasons. First, it shows a new perspective for which citizens' attitude to comply with taxes is affected by their beliefs about the well-functioning of the state. As it will be clear later, the effect that we document here goes beyond the standard channel for which citizens are more willing to pay taxes when they expect the provision of public goods and services in exchange as, in our analysis, the effect on today's intrinsic motivations of citizens comes from the expectation about *future* state capacity. Second, given the positive relationship between the citizens' intrinsic motivations to comply with taxes and the existence of an efficient enforcement system, we posit that a foreign military intervention that can credibly signal the start of a state-building process may (unintendedly) trigger a *critical juncture*, that is a path-dependent institutional trajectory that can move an economy toward a new, more efficient, institutional equilibrium (Callen, Weigel and Yuchtman, 2024).

To identify the effects of expectations about a foreign military intervention on citizens' intrinsic motivations to pay taxes, we use Mali at the juncture of 2012 as a natural experiment. Beginning with January 2012, the capability of the state of Mali to provide essential functions to its citizens weakened considerably. While the North was trapped in a bloody civil conflict, the heart of the state was shaken by a severe institutional crisis that reached its peak in the wake of a military coup in April 2012. The combination of these events left the country split and without a legitimate government. The reaction of the international community was slow but, eventually, decisive through a resolution undertaken by the United Nations Security Council (UNSC), on December 20th. The resolution explicitly authorized the deployment of force in the North of Mali and, importantly, previewed a road map of actions to take, from the defeat of the rebels to the country's reunification to the restoration of democratic elections of a new legitimate government. Hence, this announcement was likely to signal state-building by picturing a potentially different governance of Mali once the intervention was accomplished. We take advantage of this historical context to examine whether this signal of state-building raised Malians' motivation to comply with taxes.

We assemble individual information on self-reported attitudes and individual characteristics as well as detailed information on the presence of basic public goods and services at the primary sample unit (PSU) level,³ collected by Afrobarometer

³A PSU is the smallest, well-defined geographic unit for which reliable population data are

(AfB). Our analysis exploits the difference in the exposure to the UNSC resolution announcement. We compare attitudes between individuals interviewed in the days immediately before the UNSC resolution and individuals interviewed in the days immediately after. Our identification strategy relies on plausibly exogenous differences in the timing of the interviews relative to the timing of the UNSC resolution. Our regressions control for region-by-ethnic group fixed effects as well as for a large set of individual characteristics. Hence, we identify the effect on motivation to comply by comparing individuals with similar characteristics, with the same ethnic background, interviewed in the same region, in the same period, but respectively before and after the 20th of December 2012.

We first document that individuals that were interviewed the days after the UNSC resolution have characteristics that do not differ statistically from those interviewed the days before. We examine a large set of characteristics that potentially explain variation in tax compliance. We look at the gender and the age of the respondent as well as his or her employment status and educational level. We then show that the two groups of respondents are not statistically different relative to the place where they live (urban vs. rural and the number of public amenities provided), religious commitment, and belonging to the predominant ethnic group (e.g., Bambara). We also verify that interviewers were not different before and after the 20th of December 2012. This is important as interviewers are likely to affect answers in several, possible ways well documented by the theory on the social desirability bias.

Comparing these two groups, we obtain that respondents interviewed the days after the UNSC resolution are more inclined to believe that people *must* pay taxes. The effect is statistically as well as economically significant, as it explains about 0.26 standard deviations in our outcome variable. This result is corroborated by a set of regressions that use related outcomes. First, we estimate an increase of 12 percentage points in the share of citizens that would never refuse to pay taxes or fees to the government. Second, we document that the share of citizens that declare to be willing to evade if had chance—a dimension of tax compliance that captures intention to future behavior—drops by 9.5 p.p. Third, citizens interviewed the days immediately after the UNSC resolution adoption not only are more willing to pay taxes but also declare that other citizens who evade should be prosecuted: we estimate

available ([Afrobarometer, 2011](#)).

mate an increase of 5 p.p. which is however statistically imprecise. Taken together, these findings suggest that the expectation of a foreign military intervention can increase the salience of state capacity and the demand for state-building, potentially opening important windows of political opportunity for enlarging the fiscal capacity of a state.

We then explore which segment of the Malian population is particularly sensitive to the state-building signal. We find evidence that the signal intensity is only relevant in areas with a low state presence, where the expected benefits from a higher-performing state should arguably be larger (e.g., [Weigel, 2020](#)). Likewise, we obtain that the expected foreign intervention is only salient for individuals who value the state. For example, we find that those who feel discriminated by the state (and are likely to hold a negative view of it) do not experience a higher intrinsic motivation to pay taxes, and this can be interpreted as a lack of demand for state-building. Consistently with [Alesina, Baqir, and Easterly \(1999\)](#), we find that ethnic fragmentation negatively affects the support for state-building processes as only citizens in low ethnic diverse areas experience an increase in their intrinsic motivation to comply with taxes. Finally, we document that only citizens with access to news, that is with either a radio or a TV at home, react to the UNSC resolution announcement—a result which is particularly important for the credibility of our story as it provides support to one of the mechanisms behind our results that rely on the acquisition of information about the UNSC resolution by citizens.

The increase in demand for state-building goes hand-in-hand with other changes in citizens' views. We provide evidence that the shift in citizens' voluntary compliance is associated with an increase in the sense of cohesion between the citizens and the state as we find that citizens become more patriotic and develop a better judgment of the state agencies' officers, especially of tax officials. We therefore conclude that a likely mechanism driving the effect of the state-building signal on the citizens' motivations to comply with taxes is provided by their view about the state institutions, as recently stressed by [Besley and Dray \(2024\)](#). Further tests corroborate this conclusion. We show that respondents interviewed the days immediately after the UNSC resolution are more likely to believe that evading taxes is relatively more difficult; furthermore, they are more willing to resort to a state actor (e.g., police) for protection and assistance in case of need. We rather find limited evidence about other mechanisms such as the role of an expected accountability by international

Gao, Timbuktu, and part of Mopti (see Online Appendix Figure A1 that maps administrative divisions in Mali). This territory was unilaterally proclaimed as the *Azawad* state.

For months, Mali was *de facto* divided in two with the Malian army completely unprepared to face the threat and in need of reorganization. Citizens were reported losing confidence in the country's fragile state institutions. A Gallup study conducted in October/November 2012, for example, documents that Malians' confidence in the government fell by 22 percentage points (from 71% in 2011). Confidence in the military and the judiciary dropped as well by 25 and 17 points, respectively, while the perceived level of corruption rose by 10 percentage points.⁷

2.2 The intervention of the international community

The fragile Malian transition government then turned to the international community for help. However, as late as October 2012 the United Nations Security Council (UNSC) only agreed on transferring economic resources to the Malian government, while not authorizing the deployment of force.⁸ The escalation of the terror in the North, caused by the increasing leadership of Al-Qaeda warriors over the Tuaregs, induced the UNSC to urgently adopt S/RES 2085 on 20 December 2012 eventually authorizing a military intervention.⁹

The resolution was not simply an act through which the international community authorized an African Union-led mission to recover the North of Mali. In fact, military and security aspects were not mentioned before Art. 6. Essentially, it was a document through which the international community detailed a political process necessary to rebuild the Malian state: “[*The Security Council*] urges the transitional authorities of Mali [...] to finalize a transitional roadmap through broad-based and inclusive political dialogue, to fully restore constitutional order and national unity, including through the holding of peaceful, credible and inclusive presidential and legislative elections” (Art. 1). The international community was required “to assist the transitional authorities of Mali in the preparation of such a roadmap, including the conduct of an electoral process based on consensually established ground rules” (Art. 1). By the same token, the authorized African-led International Support Mis-

⁷See the Gallup study at <https://bit.ly/3BTjKhf>.

⁸See the S/RES 2017 full text (12 October 2012) at <https://bit.ly/43pLTZ3>.

⁹See the S/RES 2085 full text at <https://bit.ly/3Wv1tjN>.

in complying with taxes and is the closest to the concept of tax morale typically studied in the literature (e.g., Andreoni, Erard and Feinstein, 1998; Luttmer and Singhal, 2014). As we illustrate in Online Appendix Table A1 (where we report the main summary statistics), its average attests at 3.99 out of 5. However, a closer inspection of its distribution (see Figure A3) reveals a substantial variation around the mean (standard deviation is 1.05). We also examine other related questions. We look at the share of respondents that would never refuse to pay tax or fee to the government (i.e., the share of individuals that responded 0 to the AfB-5 question Q26C) and at the share of respondents that would evade if had chances (i.e., the share of individuals that responded 1 to the AfB-5 question Q26C).¹¹ We also employ the share of respondents who consider not paying the taxes wrong and punishable.¹² On average, 83% of the interviewees declare that would never refuse to pay taxes, 14% declared that would evade if had the chance, while 66% think that evasion must be punished.

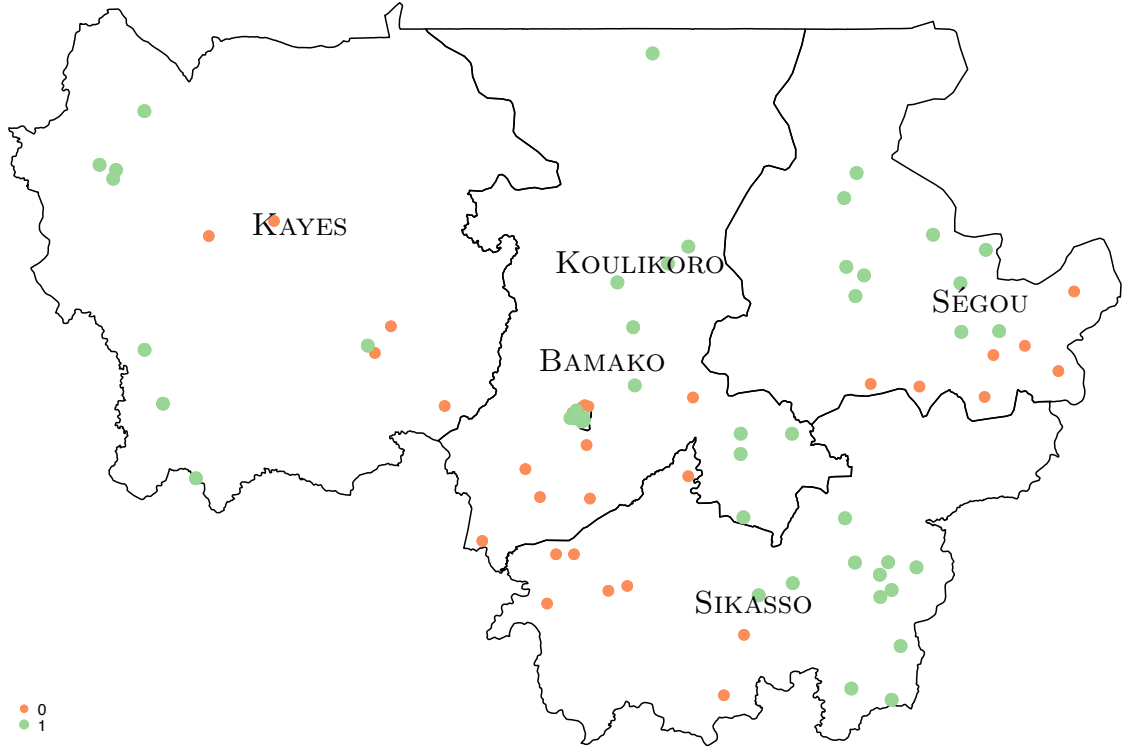
3.3 Related attitudes

To better interpret our results we employ an additional set of variables capturing a wide range of related attitudes, motivations, and beliefs. First, we explore perceived corruption. We compare shifts in generalized perception of corruption, captured by the coding of question Q60C (*“How many government officials do you think are involved in corruption, or haven’t you heard enough about them to say?”*, where 0 means *“none”* and 3 indicates *“all of them”*), and perceived corruption committed by tax officers. For the second belief we use the coding of individual responses to the question Q60F: *“How many tax officials (e.g., Ministry of Finance officials or Local Government tax collectors) do you think are involved in corruption, or haven’t you heard enough about them to say?”*, where 0 means *“none”* and 3 indicates *“all of them”*. Perceived corruption is on average higher (1.69) when referred to the entire body of public servants than to the tax department only (1.59).

¹¹Question Q26C goes as follow: *“Would you refuse to pay a tax or fee to the government if you have the chance?”* Responses are coded as 0 if the answer is *“No, would never do this,”* 1 if it is *“No, but would do if had the chance,”* 2 if *“Yes, once or twice,”* 3 if *“Yes, several times,”* and 4 if *“Yes, often.”*

¹²This is the share of respondents answering 3 to question Q76B: *“please tell me whether you think that not paying the taxes is (1) not wrong at all, (2) wrong but understandable, or (3) wrong and punishable.”*

Figure 2: Sampled villages' location and the treatment



Notes. Treated PSUs are in green, control PSUs are in orange.

4 Empirical framework

4.1 Empirical strategy

We use the timing of the survey as an instrument for exposure to the announcement of the foreign military intervention. Our instrument is therefore a binary variable, $Post^{UN}$, that switches on to 1 for respondents in PSUs sampled for interview the days after the United Nations adopted the Resolution #2085. We illustrate the empirical strategy in Figure 2 that maps PSUs using the geocoded information provided by AfB (BenYishay et al., 2017). Colors reflect the receipt of the treatment. PSUs sampled before, or on the 20th of December, are drawn in orange; those sampled the days after the 20th of December in green. We therefore estimate the difference

in tax morale, our main outcome of interest, between interviewees interviewed the days immediately after the announcement (in green) and those interviewed the days immediately before (in orange). This difference is captured by the coefficient β in the following regression:

$$y_{ijer} = \alpha + \beta Post_{ij}^{UN} + X'_{ij}\gamma_1 + R'_{ij}\gamma_2 + \Theta_{er} + \varepsilon_{ijer}, \quad (1)$$

where i indicates the respondent, j the PSU, e the belonging to a specific ethnic group,¹⁵ and r the administration region of the PSU. The vector X_{ij} includes the individual's and PSU's characteristics, that is the interviewee's age (and its square) and educational level, whether the respondent is a male, whether she/he lives in a rural area, whether she/he is not formally employed, whether she/he is a member of a religious group, whether she/he is a member of the major ethnic group (i.e., Bambara), and the number of public goods and services provided in the PSU where she/he resides (among those listed in Section 3.4). To ensure that the interviews have been conducted in exactly the same way in the two groups we include the vector R_{ij} which gathers the following interviewer's characteristics: interviewer's gender and education, whether the interviewer speaks the same respondent's language, and whether the interviewer thought anyone influenced the respondent's answers. In the most demanding estimation, we also include a set of region-by-ethnic group fixed effects, Θ_{er} , that ensure a comparison of individuals that live in the same region and have the same ethnicity. Finally, ε_{ijer} is an individual-specific error. We do not assume independence of residuals within a PSU and, consequently, we cluster standard errors at the PSU level. Later on, we will show estimates under alternate, less conservative clustering approaches.

4.2 Identifying assumption

The identification of β relies on the fact that the announcement of the military intervention splits the AfB sample in a way to generate two equally representative samples of the underlined population, thus making for a valid comparison between

¹⁵Following Eifert, Miguel and Posner (2010) and Depetris-Chauvin, Durante, and Campante (2020) we use the information on the main language spoken at home as a proxy for ethnic background. While as many as 17 languages are recorded, 65% of South Malians speak Bambara in our sample. We will show below that our results are not sensitive to using self-reported ethnic groups as an alternative.

positive statistically significant relationship between motivation to pay taxes and the expectation of the foreign military intervention. As expected, no relevant differences in our estimation emerge when we control this relationship for the respondents' characteristics (column 2) as well as for the features of the interviews (column 3). In columns 4 and 5 we add language group and region fixed effects, respectively, restricting our comparison within individuals with the same ethnic group (in column 4) and in the same region (in column 5). In column 6 we add both, while in column 7 we include language $\times$ region fixed effects. This is important as this specification allows us to hold fixed all those factors that operate within an ethnic group of a specific region that may explain variation in individual motivations to pay taxes (i.e., culture, regional laws, differences in the enforcement level, etc).

The effect is quite stable along these specifications and it is always positive and statistically significant: South Malians interviewed the days after the announcement of the UNSC resolution display higher motivations to pay taxes relative to those interviewed the days after. The effect is also economically important. Focusing on column 7, we obtain that this shift equals to 0.26, accounting for about a quarter of the standard deviation of our 1-to-5 scale definition of tax morale (recall from Table A1 that a standard deviation is 1.05).

Online Appendix Table A4 shows that our results are not sensitive to a specific way of coding tax morale (columns 2, 3, and 4) and to the specific econometric model employed in the analysis (in column 5 we use an ordered logit model to estimate our focal effect). For example, using an alternative binary definition of tax morale, that takes on the value of 1 if the respondent either agrees or strongly agrees on the statement that people must pay taxes (0 if not), we estimate that the share of interviewees with a higher level of tax morale increases by 10.8 p.p. in the post-announcement period, from a baseline of 79 p.p. (column 2 of Table A4). Table A5 additionally shows that our results are not sensitive to expanding the time window of respondents to interviews conducted up to two weeks after the UNSC resolution (column 2) and up to three weeks (column 3), the furthest horizon we can possibly use from AfB-5. Table A6 indicates that our results are robust to the use of self-declared ethnic group fixed effects in place of language group fixed effects. Finally, Table A7 suggests that the estimation precision vary imperceptibly when we use alternate clustering strategies, such as language group times region, or

in the post-UNSC resolution period relative to the pre-period—a shift that equals to around 14.5% of the sample average (recall from Table A1 that this variable has mean equals to 0.83).

In column 2 of Table 4 we look at intentions to future behavior by using a dummy which takes 1 if the respondent declares that he/she would refuse to pay tax or fee to the government if had chance. Respondents who answer yes are those with a low level of tax morale but who comply with taxes if they are forced to. Our estimate documents a difference of 9.5 p.p. between respondents interviewed in the post- and pre-period. The effect is substantial. Relative to the sample average share (equal to 0.14), the share of respondents that would refuse to pay tax or fee to the government if had chances decreases by about 72%. The effect is also conceptually important as it brings evidence that a credible signal of state-building can effectively raise voluntary compliance, an asset which is critical for countries, like Mali, with a limited capacity to tax.

Finally, in column 3 of Table 4, we explore whether, in addition to increasing individual motivations to pay taxes, the announcement of the foreign military intervention changed people’s view toward tax cheaters. We therefore look at variation in the share of respondents that believe that tax evasion is wrong and must be punished. Our estimate indicates an increase in the stigma of non-compliance. The estimated difference between the respondents interviewed in the post- and the pre-period is around 5 p.p.—which is the 8% of the sample mean. While this result is potentially of particular importance—it indicates that salient events may be effectively conducive to important fiscal reforms aimed at expanding the country’s fiscal capacity—we note that it is statistically imprecise.

7 Mechanisms

So far, we have documented that the announcement of the foreign military intervention caused a shift in people’s motivations to pay taxes. What does explain this reduced-form impact on tax morale? In this section, we bring suggestive evidence in favor of the following updating mechanism: the expectation of the foreign military intervention, aimed at reunifying the country’s territory, defeating rebels, and restoring democracy, sent a state-building signal on the future capacity of the state. The signal raised the expected benefits of participation, including paying

in a bias in our estimates. To absorb these potential effects, we use below a set of intertwined strategies.

8.1 Instrumenting news exposure using distance to Bamako

A potential violation of our identification strategy appears from a closer look at Figure 2: apparently, PSUs selected into the treatment are more distant to the capital Bamako. Despite we have checked above that treated interviewees and non-treated ones balance along a wide range of dimensions, in this section we further ask whether we might have missed an unobservable aspect that can tell something about the geographical distribution of the treatment. This is important because this unobservable can in turn be correlated with tax morale and therefore introduce an omitted variable bias in our analysis.

What can explain this pattern in the treatment? We have checked this issue with the AfB team. What we have been told is that the survey rollout from one PSU to another follows organizational and travel cost reduction strategies. Interviews started in Bamako, where AfB has its Malian headquarter, and spread across the most proximate, randomly selected PSUs (see Online Appendix Section B for details).²⁶ Hence, the most distant PSUs to Bamako have to be more likely to be treated as they were interviewed at a later date. This is indeed what we show in Online Appendix Figure A9 for the three regions (i.e. Kayes, Koulikoro, and Sikasso) that according to AfB and because of proximity to Bamako followed this strategy. We compute ventiles of the PSUs distribution according to the distance to Bamako and, for each bin, we calculate the share of PSUs interviewed the days after December 20. A strong association between the distance to Bamako and the share of treated PSUs appears in Figure A9: as we will document below (column 2 of Table 7), we estimate that a standard deviation increment in the distance to Bamako²⁷

²⁶We back-checked this conjecture on a map that we illustrate in Online Appendix Figure A8. The map displays a gradient of colors reflecting the schedule of the interviews, with darker shades indicating later dates. On top of that, we draw the geographical pattern of the interviews for each region in South Mali. In the region of Kayes, for example, interviews moved from the South-East (the closer area to Bamako) to the North-West. Similarly, in the region of Koulikoro, they proceeded northward (outbound from the capital Bamako), while in Sikasso they moved eastward starting from the northern boundary which is the closest to the capital. Because of the distance to Bamako, in the Ségou region, interviewers followed a different strategy moving from the East to the West.

²⁷The average Euclidean distance to Bamako is 1.99 (about 187 kilometers) and the standard

comply with tax laws. These events also increase the sense of cohesion between the citizens and the state, by making individuals more patriotic and with a better view of the state institutions. Our results therefore indicate that these events can create a window of opportunity to reform the state by making politically feasible investments in fiscal capacity that are typically unpopular.

We use Mali in 2012—a country almost disbanded because of a military coup and a concomitant rebellion that took control of half of its territory—as a quasi-natural experiment to test this theory. Our identification strategy takes advantage of the fact that the UNSC resolution came unexpectedly during the Afrobarometer survey rollout in the country. This gives us the opportunity to interpret causally the shift in Malians’ beliefs we document and to study a plausible critical juncture in real-time helping avoid the problem of selecting research contexts on the outcome (Callen, Weigel and Yuchtman, 2024). While specific to Mali, our research is likely to inform on the effects of other foreign military interventions aimed at rebuilding a state. Scholars of peace science have shown that peace-building operations in South Sudan (S/Res/1996) or in Liberia (S/Res/1509), among others, have been accompanied by a detailed roadmap to restore security, build capacity, and help a democratic transition (Di Salvatore et al., 2022)—all aspects that were central in the intervention in Mali.

Finally, we note that we do not observe the actual citizens’ decision to comply with tax laws—a limitation that further works should help bridge. While our study shows that the UNSC resolution announcement arguably created a critical juncture, which might have driven Mali to a better institutional equilibrium, a subsequent military intervention by France, in January 2013, overruled the UNSC decision—a circumstance which caused the abandonment of the roadmap of actions previewed in the document (e.g., Bannelier and Christakis, 2013). The institutional trajectory in Mali ultimately remained unchanged, but, as argued by Callen, Weigel and Yuchtman (2024), this does not undermine the importance of our study as critical junctures are by construction contingent with some succeeding and some failing. We see the analysis of other foreign military interventions as an important avenue of future research to inform policymakers and donors about the fully-fledged impacts of these interventions in the targeted countries.

Table 1: Balance in covariates

	(1) Difference	(2) t -statistics	(3) Normalized Difference
Respondents' characteristics			
Male	-0.001	-0.10	-0.001
Age	-0.466	-0.38	-0.022
Rural	-0.135	-1.27	-0.212
Unemployed	-0.053	-1.86	-0.105
Education	0.281	0.78	0.087
Religious Group Member	-0.034	-0.50	-0.031
Major Ethnicity	0.107	1.20	0.160
Public Goods Provided	0.408	0.65	0.109
Interviewers' characteristics			
Male	0.010	0.39	0.014
Education	-0.007	-0.31	-0.023
Same Language	-0.010	-0.18	-0.014
Influenced by Others	-0.002	-0.21	-0.014
Observations	683	683	683

Column 1 reports estimates from regressing each covariates to the dummy $Post^{UN}$, equals to 1 if the interview has taken place the days after the adoption of the UN resolution. Column 2 reports the t -statistics. Robust standard errors clustered at the PSU level in parentheses. Column 3 report normalized difference for establishing unconfoundedness (see [Imbens and Wooldridge, 2009](#)).

Table 3: Tax morale before and after the UN resolution announcement

	Dependent variable is: Tax morale						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$Post^{UN}$	0.180*	0.174**	0.183**	0.202**	0.176**	0.220**	0.263***
	(0.091)	(0.087)	(0.085)	(0.093)	(0.077)	(0.088)	(0.085)
Respondents' Characteristics	No	Yes	Yes	Yes	Yes	Yes	Yes
Interviewers' Characteristics	No	No	Yes	Yes	Yes	Yes	Yes
Language FE	No	No	No	Yes	No	Yes	No
Region FE	No	No	No	No	Yes	Yes	No
Language $\times$ Region FE	No	No	No	No	No	No	Yes
Observations	701	700	700	700	700	700	683
R^2	0.007	0.045	0.051	0.071	0.085	0.101	0.127

Dependent variable is tax morale. The unit of observation is the respondent individual. Columns 2 to 8 include age (and its square), a dummy if male, if rural, if unemployed, if a member of a religious group, if a member of the major ethnic group, the educational level, and the number of public services provided in the PSU. Columns 3 to 8 also include interviewers' characteristics, such as the gender, the education level, whether the interviewer speaks the same language as the interviewee, whether the interviewer thinks the pattern of responses has been influenced by others. Standard errors in parentheses are clustered at the PSU level. Symbols: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Other dimensions of tax morale

	Dependent variable is:		
	(1) I would never evade taxes	(2) I would evade if had chances	(3) Evasion must be punished
$Post^{UN}$	0.121*** (0.039)	-0.095*** (0.032)	0.050 (0.039)
Respondents' Characteristics	Yes	Yes	Yes
Interviewers' Characteristics	Yes	Yes	Yes
Language $\times$ Region FE	Yes	Yes	Yes
Observations	678	678	679
R^2	0.143	0.099	0.240

Dependent variable is tax morale. The unit of observation is the respondent individual. All the columns include age (and its square), a dummy if male, if rural, if unemployed, if a member of a religious group, if a member of the major ethnic group, the educational level, and the number of public services provided in the PSU. They also include interviewers' characteristics, such as gender, the education level, whether the interviewer speaks the same language as the interviewee, whether the interviewer thinks the pattern of responses has been influenced by others. Standard errors in parentheses are clustered at the PSU level. Symbols: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Tax morale before and after the UN resolution announcement: IV results based on distance to Bamako

	(1) OLS	(2) 1st stage	(3) 2nd stage	(4) Reduced form
	Tax Morale	$Post^{UN}$	Tax Morale	Tax Morale
$Post^{UN}$	0.311*** (0.106)		0.359*** (0.127)	
Distance to Bamako		0.450*** (0.050)		0.162** (0.064)
Kleibergen-Paap F-statistic		81.174		
Respondents' Characteristics	Yes	Yes	Yes	Yes
Interviewers' Characteristics	Yes	Yes	Yes	Yes
Language $\times$ Region FE	Yes	Yes	Yes	Yes
Observations	407	407	407	407
R^2	0.221	0.685	0.221	0.216

Dependent variable is tax morale. The unit of observation is the respondent individual. All columns include age (and its square), a dummy if male, if rural, if unemployed, if a member of a religious group, if a member of the major ethnic group, the educational level, and the number of public services provided in the PSU. They also include interviewers' characteristics, such as the gender, the education level, whether the interviewer speaks the same language as the interviewee, whether the interviewer thinks the pattern of responses has been influenced by others. Standard errors in parentheses are clustered at the PSU level. Symbols: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$



Impact of Village Savings and Loan Associations: Evidence from a cluster randomized trial[☆]



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ABSTRACT

The vast majority of the world's poor live in rural areas of developing countries with little access to financial services. Setting up Village Savings and Loan Associations (VSLAs) has become an increasingly widespread intervention aimed at improving local financial intermediation. Using a cluster randomized trial we investigate the impact of VSLAs in Northern Malawi over a two year period. We find evidence of positive and significant intention-to-treat effects on several outcomes, including the number of meals consumed per day, household expenditure as measured by the USAID Poverty Assessment Tool, and the number of rooms in the dwelling. This effect is linked to an increase in savings and credit obtained through the VSLAs, which has increased agricultural investments and income from small businesses.

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1. Introduction

The vast majority of the world's poor live in rural areas of developing countries. There they endure long periods of time between inputs into and outputs from agricultural production, uncertainty about harvest outcomes, and dependency on the weather. This makes it crucial that they are able to smooth consumption, to access credit and to employ strategies for coping with risk (Conning and Udry, 2007). However, the history of rural financial intermediation is not encouraging, and even the explosive global growth in microfinance has been

concentrated in urban and semi-urban areas (Allen and Panetta, 2010; Daley-Harris, 2009; Demirgüç-Kunt and Klapper, 2012).

When formal financial institutions are not available, households rely instead on networks, moneylenders and informal financial mechanisms, which exist in a myriad of forms (Collins et al., 2009; Rutherford, 2001). Some of the most well-known are perhaps ROSCAs (rotating savings and credit associations) (Besley et al., 1994).¹ In its simplest form, members contribute to a common pot of money that is awarded to a different member at each meeting. Apart from the first and last members, each member will be both a saver and a borrower in the course of a cycle (Anderson and Baland, 2002; Besley et al., 1994; Bouman, 1995; Klonner, 2008). Village Savings and Loan Associations (VSLAs) – a type of local financial institution that has been standardized by CARE and is increasingly being promoted by a host of different organizations – take many of the elements of ROSCAs and add more flexibility in savings and loans, standardize the governance structure and reinforce accountability elements. Each VSLA relies on its members'

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¹ Other examples are ASCAs (accumulating savings and credit associations), and Susu collectors originating from Ghana ('Susu' meaning 'small small' in the Akan language in Ghana).

savings to provide credit (as well as small contributions to a limited fund which can be accessed in times of funerals or illnesses of household members). A minimum level of savings is compulsory at the weekly meetings. Typically once a month loans are made available to members from the VSLAs, and a stated aim of many VSLA projects is to encourage small businesses. VSLAs thus create a local financial market that allocates local savings to debtors who are group members. Moreover, VSLAs implement a number of accountability and governance features that are described in more detail below. These features lead to low cases of default or elite capture.

Once a year, all the savings and the accumulated interest are paid out. In our context, rural Malawi, a large number of groups plan this “share-out” just before the agricultural planting season, when seeds and fertilizer are bought. This also coincides with the beginning of the school year when education related expenses are high. There are few formal alternatives for saving and receiving credit in the villages we are studying: at baseline only 0.6% of households are participants in a formal micro-finance institution, and 5 to 6% in a VSLA group for example.

VSLAs have now been introduced in 72 countries and have 11 million active participants worldwide.² However, despite this wide distribution and the increasing popularity among donors as a means of improving the financial infrastructure in remote rural areas, very little is known about the impact of VSLAs on household welfare. Unlike other types of microfinance for which several randomized impact studies have been made showing limited effects (see Banerjee (2013) for a review of microfinance, and Banerjee et al. (2015c) for a review of microcredit), rigorous impact evaluations of VSLAs that address endogenous program placement and selection bias have only recently been completed.³

We provide a rigorous impact assessment of VSLAs on pre-defined household outcomes. We do this through a cluster randomized control trial carried out in forty-six villages in northern Malawi. Among these forty-six villages, twenty-three villages were randomly selected to participate in a VSLA project implemented by a local NGO, Soldev, from 2009 onwards. The remaining twenty-three villages were used as a control group, before the project was finally extended to them in 2011. We conducted household surveys of 1775 households in 2009, just before the project was implemented in the treatment villages, and again in 2011, just before the intervention was introduced in the control villages. We carefully tracked households that moved between the 2009 and 2011 surveys, resulting in a low attrition rate of only 3%. By 2011, 45% of the households in treatment villages had joined VSLAs, while almost 21% of households in control villages had also joined. Despite the implementing NGO adhering strictly to the randomization design, there was spontaneous spill-over into control villages – typically because control village inhabitants decided to join VSLA groups in treatment villages.

We assess the impact of introducing VSLAs at village level on seven general indicators of household welfare. These were pre-defined as targets by the implementing partner as part of the project design. We operationalize the seven indicators by one or two outcome variables each (see Table 2) leading to a total of nine final outcome variables. We also provide multiple hypothesis corrected p-values (denoted by q in the following).

We find evidence of positive impacts on four out of nine outcome variables. Food security was improved in that the number of meals consumed per day increased by 0.13 (s.e. = 0.05, $p = 0.02$, $q = 0.09$)⁴,

corresponding to one more meal per week. Households were found to hold larger savings in total ($p = 0.02$, $q = 0.07$). Increases in households' scores on the USAID's Poverty Assessment Tool (PAT)⁵ suggest that expenditures increased by about 4.2% (s.e. = 0.02, $p = 0.05$, $q = 0.11$). Finally, the average number of rooms per dwelling rose by 0.16 (s.e. = 0.06, $p = 0.02$, $q = 0.09$). Concurrently, there is some evidence that the number of income generating activities decreased, which could be consistent with increased specialization.

Overall, the results provide evidence that VSLAs can improve household outcomes, even in as short a period as two years, without any injection of outside capital. The impacts on participating households are likely to be larger, as our impact estimates are given by the intention-to-treat estimator, i.e. they are average impacts across both participating and non-participating households. Furthermore, there has been a non-negligible degree of spill-over into control villages and we may therefore be underestimating the actual impact size. The results of positive impacts are consistent with results in Beaman et al. (2014), who study VSLA groups in Mali. They also report positive impacts on savings (in particular livestock) and food security.⁶

In the literature, there are several suggestions for possible channels through which improved financial intermediation and participation in savings groups can have a positive impact on the participating households and local communities. For VSLAs to attract members, they need to offer higher interest rates or more secure repayments than alternative forms of savings. This is possible if groups are better at monitoring loans than individuals are, if groups have stronger enforcement mechanisms than individuals do, or simply because groups pool the risk on individual loans (Diamond, 1984).⁷ In turn, the larger pool of savings coupled with better loan monitoring (and presumably lower interest rates on loans than those offered by money-lenders) expands access to credit within each VSLA, though only to the extent that savings have accumulated.^{8,9} If households are constrained in their access to

⁵ The USAID Poverty Assessment Tool, usually abbreviated as PAT, is a list of twenty simple indicators that are chosen so as to measure poverty in expenditure as accurately as possible around the 1.25 USD a day poverty lines (see USAID, 2009 and www.povertytools.org/countries/Malawi/Malawi.html). Indicators included in the PAT for Malawi are chosen based on the ability to predict household poverty by predicting household expenditure using the Second Integrated Household Survey (a survey in the World Bank's Living Standards' Measurement Surveys series). The indicators include household member characteristics (the age and marital status of the household head, the size of the household, literacy and educational level of household members), household characteristics (the number of rooms, the type of floor, the type of toilet facility as well as electricity in the household), assets (ownership of bed, iron, radio, and bicycle respectively as well as raising or owning livestock in general and goats in particular) as well as other indicators (whether the household purchased any soap in the past month and whether the household grows a vegetable garden). The typical use of the tool is to measure average poverty status in a population. The calculation of poverty status involves as a first step predicting expenditure levels. We use the estimated expenditure levels directly, noting that the predictions are most accurate around the 1.25 USD poor/non-poor cutoff. Since the average (predicted) level of expenditures at baseline is around 1.17 USD per person per day, the prediction is likely to be reasonable good. While there is considerable documentation on how good PAT is at measuring expenditure-based assessments of poverty, as of now, there is no evidence on how good it is at measuring changes. It is likely that the measurement error introduced by the prediction would reduce the power to detect effects.

⁶ Annan et al. (2013) document substantial effects of the VSLA program, though they note that their outcomes are mostly based on self-reports.

⁷ By reducing the risk involved in savings, the not quite “risk-free” but lowest-risk option can also enable households to adjust their portfolio of activities and undertake more risky investments elsewhere, for example in businesses or agriculture.

⁸ As such, this is quite different from a micro-finance institution expanding access in villages and providing new resources for credit, as studied in Fulford (2013). He finds that when household have precautionary saving motives and gain access to credit, there can be an initial consumption boom at the expense of precautionary savings. In VSLAs, this cannot be true on average.

⁹ In addition, VSLAs typically offer a simple and small insurance component. The groups themselves decide on the exact nature of the insurance, but it almost always involves insurance against the illness or death of household members. That is, the insurance might be an additional device for coping with risk that can encourage households to discard inefficient ex-ante coping strategies, such as low risk/low return activities. If households choose activities involving a higher level of risk but higher expected returns, this can lead to increased consumption in the long run even if the insurance itself never has to pay out (Carter and Zimmerman, 2003; Dercon, 1996).

² Based on information on the website www.vsla.net, maintained by VSL Associates, an organization that provides training in setting up VSLAs (site accessed on September 9th 2015).

³ We know of a number of project reports that report on VSLA interventions. Three of these use randomized control trials. Annan et al. (2013) is an evaluation report on a study of VSLAs in Burundi; Beaman et al. (2014) study VSLAs in Mali. A larger international study covers Ghana, Malawi and Uganda, but at the time of writing, is not yet available for citation.

⁴ We present the multiple-hypothesis corrected equivalent of p-values, the sharpened q-values (Anderson, 2008; Benjamini et al., 2006) alongside the p-values. See the methodology section below.

credit for profitable business purposes, an expansion in access to credit increases household income.

Furthermore, since VSLAs entail a minimum level of compulsory savings, participants with time-inconsistent preferences who realize their self-control problems may use VSLAs as a commitment device (Ashraf et al., 2006; O'Donoghue and Rabin, 1999). Even if participants do not have time-inconsistent preferences, VSLAs might allow them to commit the rest of their households in accordance with their own preferences (Anderson and Baland, 2002; Jackson and Yariv, 2014).¹⁰ VSLAs disburse savings once a year, and typically time these share-outs to coincide with periods when most households need lump sums for investments in agriculture and education related expenses, facilitating such commitments.¹¹

Although we do not aim to investigate all of these possible channels through which the observed impacts may have come about, we do investigate the impact on economic activities that VSLAs are meant to enable, i.e. where implementing NGOs think that there are large potential returns that households do not reap because of credit constraints. Beyond the increase in total savings, we find significant increases in the use of credit, including credit used for investment purposes. Likewise, respondents report that they use their savings upon share-out for investments, primarily in agriculture and business. We follow this flow of money into agriculture and small-scale businesses, finding a statistically significant increase in the use of fertilizers for maize. There is an increase in the value of maize sold, though no increase in maize production overall. We also find a significant increase in the total income from enterprises. Overall, the main channel through which the observed positive impacts on consumption seem to occur is through providing households with a means to save for investments, with the investments into businesses paying off.

2. The intervention and research design

2.1. The VSLA intervention

The VSLA intervention encourages the formation of savings groups with typically fifteen to twenty-five members each, and trains each group in the management of their own village savings and loan association. As no external capital is provided, the groups are essentially self-managed financial intermediaries.

Within the larger microfinance sector, community-managed microfinance belongs to the category of member-based, community-managed, accumulating microfinance institutions (see Table 1). Typically, microfinance impact evaluations have focused on professionally-managed microfinance institutions. Unlike such institutions, VSLAs do not depend on the injection of external funds, just as they do not depend upon the sustainability of a professionally-managed institution, but rather on the sustainability of the group formed within the local community.

The inspiration for VSLAs came from rotating savings and credit associations (ROSCAs) and was developed by CARE international and VSL Associates during the 1990s (Ashe, 2002). The aim was to improve

on ROSCAs in two respects, namely by making the groups more sustainable and more flexible. Increased sustainability comes from a series of accountability features that prevent both the theft of funds and elite capture. Flexibility is increased in that members can at any time borrow any amount they wish up to three times the level of their own savings, provided that funds are available. Whereas ROSCAs multiply without external facilitation, VSLAs only do so to a much lesser extent, and thus require facilitation by, for instance, an NGO in order to spread rapidly. This is perhaps due to the rather complex governance features. VSLAs are implemented as described in Allen and Staehle (2007). After conducting awareness meetings in every targeted village, a local NGO facilitates the formation and training of groups. Participation in groups is free.¹² Initially, groups are visited every week in the first three months to set up the procedures. Groups work as member-owned financial intermediaries with three products: savings, credit and a smaller insurance component. Savings are compulsory; the amounts are collected at the weekly meetings and are conceptualized as buying shares. Every week, a member must buy at least one share and is permitted to buy up to five. The share value is set by the group and written into the group's constitution. It varied between MWK 50 and 100 in the groups in our study.¹³

Loans are available at every fourth meeting. If the funds requested by members exceed the amount of funds saved, the group decides who will be given loans by following a predetermined list of criteria written into the group's constitution. This typically assigns funds on the basis of the stated planned use of the loan. The interest rate on loans is set by the group, and can thus be used to regulate excess supply or excess demand in the medium term, making VSLAs less flexible in that regard than "bidding ROSCAs", which use a process of bidding for loans (Klonner, 2008). Usually, the nominal interest rate on loans is set at between 5% and 20% per month, but extensions in repayment schedules and inflation make the real interest rate considerably lower (Rasmussen, 2012). Loan contracts run for three months, with a grace period of one month. Rules for loan approval are set down in the group's constitution, but often focus on the requirement that the money be used for productive (income-generating) uses.

At the end of a cycle, which is around 12 months in duration, all savings and interest payments are divided by the number of shares and paid out – the "share-out". The average value of savings during the first share-out from all groups in the area was 6367 Kwacha (or 69.27 USD PPP-adjusted) which earned an average of 1801 MWK (19.59 USD PPP) in interest. The interest rate on savings typically works out at 4% to 5% per month, or about 60% per year (Rasmussen, 2012).¹⁴

The actual date of the final share-out is set by the group and tends to be chosen according to when most households need cash. In Malawi, this is most frequently at the end of the calendar year, when seeds and fertilizer need to be financed, as well as educational materials purchased. The NGO highlighted the sharing-out at the end of the year as a benefit from the participation in VSLAs.

At the end of a cycle, members decide whether to leave or remain in the VSLA group, and whether the group should accept new members. Any impact found in our analysis below will thus be the impact of at most two full cycles of collecting savings, distributing loans and returning the savings with interest among the 100 groups established by mid-2011. Usually after one year of initial training and monitoring, groups "graduate", which means that they cease to be supervised by the NGO that helped set them up.

¹⁰ Indeed, Jackson and Yariv (2014) show that, even when an individual's preferences are not time-inconsistent, all households will exhibit time-inconsistent preferences, unless discount rates are equal across household members or decisions are dictatorial (i.e. one household member's preferences decide). Cultural norms would suggest that this would usually be the husband's preferences, which might be another reason for most VSLA members being women.

¹¹ If, for example, the share-out is at the beginning of the agricultural season, investment in agriculture might increase. Duflo et al. (2011) find that providing farmers with an opportunity to pay for fertilizer just after the harvest, when money is available, and having it delivered at the time of planting, increases fertilizer use dramatically. In addition, the timing of the share-out might serve as a "label", fixing the minds of the participants on using the money from the share-out on a particular asset – in this case, fertilizer or other lump-sum investments in agriculture (Thaler, 1990). These increased investments in, for example, agriculture can increase productivity and in turn lead to higher household income and an increase in household welfare.

¹² Greaney et al. (2013) provide evidence that charging for participation might lead to improved outcomes in a similar type of savings group, as instructors have an incentive to provide better training.

¹³ Throughout the paper we report monetary values in Malawian Kwacha (MWK). In 2009, 1 USD corresponded to 91.91 MWK using poverty-adjusted PPP exchange rates in 2009.

¹⁴ It is lower than the interest on loans primarily due to the facts that not all the funds are lent out all the time and that savings accumulate over time.

Table 1
Microfinance overview.

Ownership	Management	Fund Accumulation	Examples
Member-based institutions	Professionally-managed microfinance	Large accumulating savings and credit associations (ASCAs) Rotating savings and credit associations (ROSCAs)	FECECAM, Benin Savings and Credit Cooperatives (SACCOS)
	Community-managed microfinance	Small accumulating savings and credit associations (ASCAs)	Tontines, susu, upatu, merry-go-rounds, chit, pasanakus
For-profit institutions	Professionally managed		Village savings and loan associations, Savings for Change, the WORTH model, savings and internal lending communities FINADEV, Benin Equity Bank, Kenya SKS, India
Non-profit institutions	Professionally managed		Grameen Bank, Bangladesh BRAC, Bangladesh Opportunity International

Apart from savings and loans, VSLAs also offer a smaller insurance component, financed by a small premium paid by each member each week separately from the savings and credit activities. The insurance is paid out as a transfer or an interest-free loan when certain events occur that are outlined in the constitution, usually the death of family members, death of cattle, sudden illness, or other emergencies.

A number of governance features guard against theft and elite capture. With the help of the facilitator, each group develops a constitution that describes areas of potential conflict and their solution, for example lending rules, election procedures, exclusion of members, and fines for delays and non-attendance. To achieve transparency in transactions, all transactions take place in the presence of all members at the weekly meetings, and funds are counted independently and in public by two elected money counters. Between meetings, funds are stored in a cashbox, locked with three padlocks. Three different members each hold a key to the box and the box is stored in the house of a fourth, so that the possibility of collusion for purposes of theft is greatly reduced. The identity of the “guardian” is kept secret from people outside the group, providing a degree of protection against theft by outsiders.

2.2. The design of the experiment

The crucial challenge for an impact evaluation is to construct a credible counterfactual that is not sensitive to selection bias, arising typically due to non-random program placement and self-selection into program participation (Angrist and Pischke, 2009; Banerjee and Duflo, 2009; Duflo et al., 2007).

We addressed the problem of non-random program placement by randomizing the roll-out of the VSLA intervention at the village level, i.e. a cluster randomized control trial. Out of 46 villages in the program area, we randomly chose 23 villages for implementation in the first year (the treatment villages) and 23 villages for implementation in the third year (the control villages). To improve the balance in observable and unobservable characteristics we grouped villages into strata based on certain characteristics believed to be correlated with central outcomes of interest (Bruhn and McKenzie, 2009).¹⁵

The randomization was carried out under our supervision by field officers from the NGO, who drew village names from seven hats containing the villages in each stratum. Fig. 1 below shows the physical location of the village centers, with the shape of the inner symbol indicating the stratum to which the village belongs, and the outer symbol indicating allocation to the treatment or control group, respectively circle and square.

As part of the VSLA intervention, the implementing partner carried out awareness meetings in *all* 46 villages prior to randomization. A more detailed description of the project introduction activities is contained in Appendix A. We used these awareness meetings to have field officers from the implementing partner collect lists with the names of the villagers who expressed an interest in joining the VSLA groups to allow for stratification on initial interest. Fig. 2 provides an overview of the timing of the randomization, the implementation and the data collection.

The unusual element of asking respondents both in the treatment and control group whether they were interested was done to increase power by oversampling respondents more likely to participate since take-up is often found to be low in microfinance projects.¹⁶ Appendix Table A1 shows that the being on the initial list of interested households does increase the participation probability as expected. However, households not on this list also had substantial participation rates. In the control villages, the probability of participation is 15 percentage points higher for those households on the initial list of interested household, while in treatment villages being on the list raises participation by 25 percentage points relative to households not on the list of initially interested households.

There are two major downsides to this design: First, since oversampling necessitates weighted regressions, a mechanical consequence is that the gains of over-sampling are not as large as one might think, in particular since participation rates among those *not* on the list of interested households turned out to be quite high, between 35 and 40%. Second, asking the control group whether they are interested can alter the behavior of the control group and can lead to ethical issues unless combined with a roll-out design (i.e. the control group will eventually also be treated).

As a consequence, the treatment effect could be biased if the control group alters its behavior in anticipation of later participation in the treatment.¹⁷ In our case, households in control villages may wait to invest in businesses, hoping to get better interest rates on credit once members of a VSLA; disinvest from their buffer stock savings in the hope that the insurance mechanism from the group they might join in the future would cover them; or the initial awareness meetings in all villages may have inspired households from control villages to seek participation in VSLA groups in treatment villages. In fact, more than 20% of households in control villages have become members of a VSLA before the roll-out into control villages started in 2011, see Table 4. This further reduces the power that was to be gained from oversampling interested households. In hindsight, the range of problems the announcement of the program might bring, in particular to such a complex topic as financial markets, suggests that it should be used with care.

¹⁵ The non-overlapping strata were defined as follows: large fishing villages, small fishing villages, particularly eager villages (identified by the field officers based on the reaction from villagers at the awareness meetings), large non-fishing villages, villages with a rice irrigation scheme, villages with another NGO-led intervention, and a final group of the remaining villages.

¹⁶ See for example Banerjee et al. (2015a) for an evaluation of a microfinance project that led to an increase in microcredit of 8.5%.

¹⁷ This is slightly different from the John Henry effect which refers to the control group trying to prove themselves by over-performing. Here they could also be thought to “under-perform”, by not investing for example.

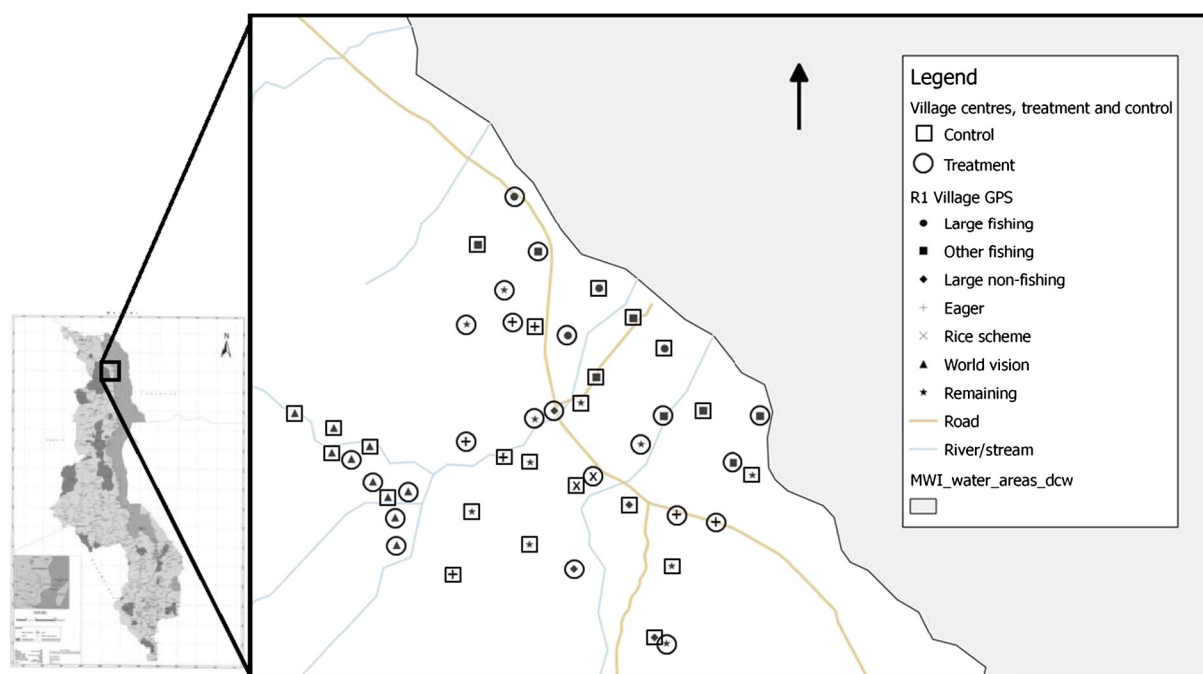


Fig. 1. Map of treatment and control villages in randomization blocks.

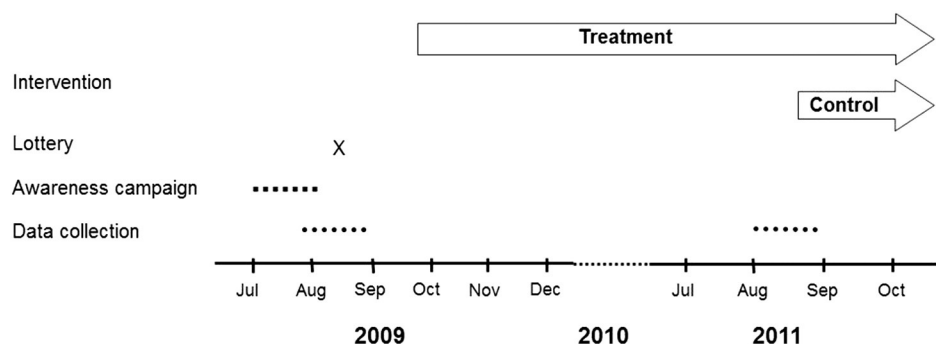


Fig. 2. Timeline of intervention and data collection.

2.3. Outcomes of interest

The final outcome variables for the primary impact analysis were closely linked to the development outcomes originally stated by the implementing NGO in the project's logical framework analysis (LFA) matrix, in which the NGO described the desired and anticipated outcomes and associated assumptions.¹⁸ The matrix shows indicators on which the intervention is expected to have an impact, but it does not specify exactly how each of these indicators is to be measured. We select final outcome variables which come close to the original specification (see Table 2).¹⁹ Although this is not equivalent to having laid out a pre-analysis plan as suggested by Casey et al. (2012), we follow this strategy to effectively tie the analysis – and our hands – to these outcome measures.

We do, however, also report extensively on non-predefined outcomes in order to investigate possible channels of the observed impact. Below we explicitly distinguish between the predefined primary

analyses following the LFA indicators and the non-predefined secondary analyses investigating mechanisms.

3. Data and estimation strategy

3.1. Data

To enable oversampling of households which had expressed an interest in participating in the groups to be formed, we stratified the sample within each village by whether the household had declared an interest in participation, using the information gathered by the NGO during the awareness meetings. We sampled roughly 38 households in total in each village.²⁰

¹⁸ The logical framework matrix is a part of the logical framework analysis (LFA), by far the most common project management tool used in international development (Dale, 2003). The matrix is made by the implementing partner prior to initiating a project and specifies the expected impacts of the intervention. Since the matrix is made by practitioners rather than researchers, it is not very detailed when it comes to measurement and analysis.

¹⁹ For a variety of reasons we did not collect information necessary to compute a total yearly household income measure.

²⁰ Due to differences in village size, our procedure led to considerable variation in sampling probability between observations, and we used weighted regressions throughout. There is no agreement on whether to use sampling weights or not (see Angrist and Pischke, 2009). If the population regression function is interpreted as causal in all its parameters, as would be the case in structural equation modeling, then weighting is in some cases not justified, regardless of the underlying method of sampling (Cameron and Trivedi, 2005). If, however, there is only one parameter of interest, for example because the causal interpretation stems from exogenous variation rather than from a correct structural specification, weights should be applied. This is the case within a counterfactual framework, or more specifically the Rubin causal model (Angrist and Pischke, 2009; Rubin, 1974).

Table 2
Predefined outcome measures and corresponding outcomes for the analysis.

Logical framework indicator (Soldev's choice)	Corresponding final outcome variable (authors' choice)
Hungry period is reduced	1) Number of months in past year with less than three meals per day.
Increase in the consumption of food	2) Number of meals previous day. 3) Food consumption per week per adult equivalent (log)
The average number of IGAs carried out by the VSLA participants has increased	4) Number of income-generating activities in each household.
Increase in the volume of savings by the VSLA groups from project-related activities by 2012	5) Total savings (log)
The share of the targeted population living below USD 1.25/day has decreased as measured using USAID's PAT	6) USAID PAT's prediction of per capita expenditure (log)
HHs have improved their housing standards	7) Number of rooms in dwelling 8) House has cement floor
Increase in household asset ownership	9) Asset count

Data were collected in 2009 and 2011 by the Invest in Knowledge Initiative under the supervision of the authors. In 2009, data collection took place from July 26 to August 30. The endline data was collected between July 8 and August 14, 2011. In both the 2009 and 2011 data collection, 24 interviewers completed interviews with up to 1775 households. A sub-sample of 834 households was also interviewed using a longer questionnaire that went into greater depth, in particular with respect to more detailed questions on loans and business activities. Fig. 1 shows the geographical locations of the villages in which the project was implemented surveyed. As can be seen from the figure, the treatment villages (marked by circles) and control villages (marked by squares) are contiguous, which made it possible for some households from control villages to participate in VSLA groups in the treatment villages, an issue which we address below.

To limit attrition, we implemented a tracking survey in 2011, with the result that a total of 1715 households were surveyed in that year. The attrition between the 2009 and the 2011 survey rounds was only 60 households, or less than 4% of the initial sample, which compares favorably with other panel surveys (Glewwe and Jacoby, 2000).

Since we also tracked households where one of the designated respondents had moved out due to some reason such as divorce, we have a number of split households, i.e. instances where one household from the 2009 survey had become two households in 2011. In the estimations below we drop one of these new households at random to ensure a balanced panel. We assess the implications of this for the results in the robustness section by limiting the sample to intact households.

3.2. Baseline balance

We test whether the randomization of villages into treatment and control groups did in fact lead to two groups with the same observable characteristics.

Table 3 below provides descriptive statistics for our final outcome variables and some characteristics of the households for the entire sample of households in 2009. Men headed 85% of households in the sample. Household heads on average had just below seven years of education. Households had an average of just below six household members each who, at the time of the interview, had a level of expenditures of approximately USD 1.17 per person per day (2005 values) assessed using the USAID PAT measurement,²¹ and ate fewer than three meals a day for an average of more than four months of the year.

Columns four and five report the mean values of the variables for the households in the treatment villages and control villages respectively. Column six shows t-values for the differences between treatment and control villages, indicating whether the differences are statistically significant. These t-values were calculated by implementing an OLS regression which controls for randomization strata while using sampling weights and clustering at the village level. None of our predefined outcomes or observable household characteristics are significantly different at a conventional 5% level and the treatment and control samples appear reasonably balanced. Only the number of income generating activities is slightly lower in the treatment than in the control villages, which is significant at the 10% level of significance. It should be noted, that total savings were only collected for the subset of households, which were administered the longer in-depth questionnaire mentioned above.

3.3. Empirical strategy

The randomization allows for simple estimation strategies to be used to estimate the treatment effects of introducing VSLAs to villages, as it ensures that without treatment, the expected outcomes are the same in the treatment villages and in the control villages. Several different estimators can be employed to estimate the intention-to-treat effect.

We first estimate the intention-to-treat (ITT) effect, δ^{DM} , i.e. the average effect of introducing the VSLA intervention for all the households in the treatment villages compared to all the households in the control villages, irrespective of whether the households actually participated in VSLAs as follows:²²

$$y_{ij} = \alpha + \delta^{DM}VSLA_j + \theta Block_j + \varepsilon_{ij}$$

where y_{ij} is the outcome for household i in village j , $VSLA_j$ is a dummy indicating whether village j was assigned to participate in the VSLA program, and δ^{DM} is the difference-in-means ITT estimate.

We include the randomization strata, termed blocks, in all our linear regressions, as recommended by Duflo et al. (2007). We implement weighted regressions using the inverse sampling probability as the weight for each household, and cluster standard errors at the village level. These sampling weights account for the fact that we over-sampled interested households.

Our second specification includes the lagged value of the dependent variable.

$$y_{ij} = \alpha + \delta^{DMlag}VSLA_j + \tau y_{ij-1} + \theta Block_j + \varepsilon_{ij}$$

Third, we use the pooled difference-in-differences estimator (Angrist and Pischke, 2009):

$$y_{ijt} = \alpha + \beta VSLA_j + \gamma post_t + \delta^{DiD}(VSLA_j \cdot post_t)_{jt} + \theta Block_j + \varepsilon_{ijt}$$

where $post_t$ indicates whether the observation is from the 2011 post-treatment survey, and $(VSLA_j \cdot post_t)_{jt}$ is the interaction of the two — i.e. whether the observation is from a treatment village in the 2011 survey. δ^{DiD} is our parameter of interest — the ITT using pooled difference-in-differences.

Finally, we take out any time-invariant unobserved heterogeneity at the household level through first-differencing.

$$\Delta y_{ijt} = \alpha + \delta^{FD}VSLA_j + \Delta \varepsilon_{ijt}$$

δ^{FD} gives us the ITT estimate using first differences, which is our preferred estimate for uncensored outcomes.

²¹ Throughout the analysis, we used the 2005 poverty-adjusted purchase power parity exchange rates suggested by Deaton and Dupriez (2011).

²² Note that because the compliance in control villages was imperfect, this will underestimate the impact of VSLAs.

Table 3
Baseline characteristics and balance between treatment and control groups.

Variable	(1) N	(2) Mean	(3) SD	(4) Treatment average	(5) Control average	(6) Difference (t-value)
Project outcomes						
Number of months with fewer than three meals a day	1737	4.10	4.03	4.26	3.92	1.18
Number of meals yesterday	1737	2.65	0.56	2.61	2.70	1.45
17-Food consumption per week per adult equivalent (MK, log)	1737	6.10	0.58	6.07	6.14	1.52
Number of income-generating activities (including agriculture and livestock)	1737	1.99	1.10	1.94	2.04	1.73*
Total savings (log)	576	7.87	1.66	7.77	8.00	0.59
Per capita expenditure predicted by USAID PAT (log)	1737	0.16	0.42	0.16	0.16	0.39
Size of house (number of rooms)	1737	2.75	1.25	2.73	2.77	0.37
House has cement floor	1737	0.10	0.30	0.11	0.09	1.47
Asset count	1737	12.89	7.87	12.84	12.94	0.46
Other household characteristics						
Age of household head	1726	38.94	15.34	39.00	38.88	0.15
Household size	1737	5.77	2.46	5.71	5.83	0.68
Household is female-headed	1737	0.15	0.36	0.17	0.14	0.87
Years of education in household	1734	6.87	3.26	7.07	6.65	1.61
Household owns land	1732	0.96	0.19	0.95	0.97	1.48
Household is member of VSLA	1734	0.06	0.23	0.06	0.05	0.66

Notes: Table presents the mean and standard deviation of baseline characteristics in the sample as a whole (columns 2 and 3), as well as the mean separately by treatment status in columns 4 and 5. Displayed results are on the full sample of 1737 households for which we have information for all pre-defined outcomes. Total savings information was only collected from the subset of 834 households, which were administered the longer questionnaire. Log (savings) only presented for observations with non-zero savings. Observation numbers are reduced when there is missing information in a central variable. Column 6 reports the absolute value of the t-statistic from a weighted regression of the dependent variable on an indicator variable for VSLA and stratification fixed effects, testing for baseline difference in means between treatment and control villages. Huber–White standard errors clustered at the village level presented in parentheses. * $p < 0.1$.

3.4. Non-linear models

We implement tobit regressions for censored variables on a logarithmic scale (Cameron and Trivedi, 2005). As it is not obvious how to provide the correct estimate and inference in difference-in-differences and difference-in-means with lagged variable tobit models, we focus on difference-in-means models for variables on a logarithmic scale.²³

When we investigate impacts on the profitability of businesses, we implement median regressions for business income, as we have a large proportion of zeros (suggesting a censored model). However, tobit regressions are inappropriate as business incomes can be negative. The OLS estimates are sensitive to outliers, resulting in large standard errors. The quantile regressions are robust to the large positive and negative values we observe, and are useful at establishing whether there is any impact of VSLAs on business income.

3.5. Multiple hypothesis testing

When multiple hypotheses are tested, the probability of rejecting at least one null hypothesis is higher than the significance level, even when a program has no effect on any of the outcomes (see, for example, Anderson, 2008). To deal with this issue, a number of techniques have been developed, the most famous of which is the so-called Bonferroni correction. Bonferroni's correction is however excessively conservative, leading to a substantial loss of power. Anderson (2008) provides an overview and implements a range of methods to correct standard errors for multiple hypothesis testing, including controlling for the false discovery rate (Benjamini and Hochberg, 1995), a procedure recently used by Banerjee et al. (2015b). We present the sharpened q-values of Benjamini et al. (2006) and Anderson (2008) for our main results, see Appendix B for more detail.

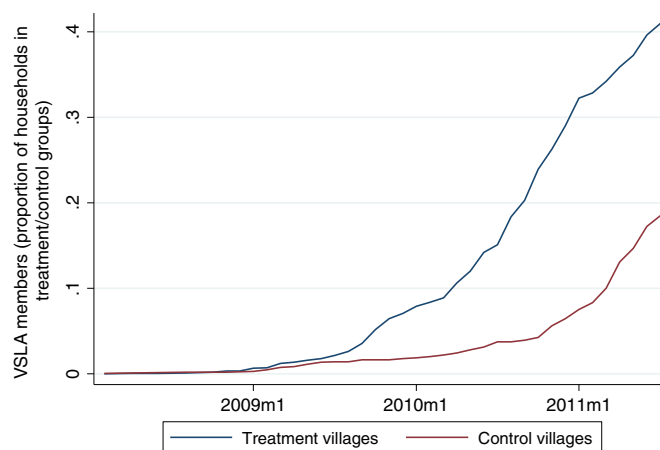


Fig. 3. Cumulative VSLA membership by village category. Notes: This figure presents the timing of membership in VSLA in treatment and control villages. Sample observations are reweighted so that they represent (estimates of) the proportion of households. Information on the timing of membership is based on survey data.

4. Results for main outcomes

4.1. Take-up

Before assessing the impact of the intervention, we describe how successful it was in attracting participants to form VSLAs. Fig. 3 gives

Table 4
VSLA membership.

	Control villages	Treatment villages	Differences
Baseline (2009)	0.052	0.061	0.010
Endline (2011)	0.207	0.451	0.259***
Difference	0.155***	0.390***	0.235**

Notes: Standard errors used in calculating significance are clustered at the village level. ** $p < 0.05$, *** $p < 0.01$.

²³ See Ai and Norton (2003) and Puhani (2012). The Stata program for the fixed effects tobit estimator based on Honoré (1992) did not converge in our case.

Table 5
Effects on predefined outcomes.

Outcome	(1) Difference in means	(2) Difference in means with lag	(3) Difference-in-difference	(4) First-difference
Number of months with fewer than three meals a day	−0.203 (0.38) [0.703] 0.055 (0.03)	−0.272 (0.36) [0.35] 0.069** (0.03)	−0.493 (0.45) [0.303] 0.145** (0.07)	−0.555 (0.38) [0.181] 0.126** (0.05)
Number of meals yesterday	[0.378]	[0.121]	[0.252]	[0.091]
17-Food consumption per week per adult equivalent (MK, log)	0.026 (0.05) [0.703] −0.179***	0.046 (0.05) [0.326] −0.144***	0.080 (0.07) [0.303] −0.104	0.097 (0.06) [0.181] −0.060
Number of income-generating activities (including agriculture and livestock)	(0.06) [0.034] 1.034** (0.43)	(0.05) [0.063] <i>n.a.</i>	(0.11) [0.338] <i>n.a.</i>	(0.07) [0.324] <i>n.a.</i>
Total savings (log)	[0.07]			
Per capita expenditure predicted by USAID PAT (log)	0.030 (0.03) [0.53] 0.104 (0.10) [0.53]	0.037* (0.02) [0.121] 0.133* (0.07) [0.121]	0.041* (0.02) [0.252] 0.135* (0.07) [0.252]	0.042** (0.02) [0.107] 0.158** (0.06) [0.091]
Size of house (number of rooms)	0.019 (0.02) [0.53]	0.002 (0.02) [0.61]	0.001 (0.02) [0.593]	−0.010 (0.02) [0.324]
House has cement floor	−0.643 (0.62) [0.53]	−0.455 (0.38) [0.234]	−0.484 (0.42) [0.303]	−0.374 (0.35) [0.324]
Asset count	[0.53]	[0.234]	[0.303]	[0.324]

Notes: Table presents results for the ITT impact of VSLA on predefined outcomes. With the exception of row 5 (total savings) all regressions are weighted least squares regressions. For total savings, weighted tobit regressions are reported, which is why we do not report results in columns (2)–(4). All columns report the estimates from a regression of the respective predefined outcome on a VSLA treatment assignment dummy and stratification fixed effects, accounting for sampling weights. Huber–White standard errors clustered at the village level are presented in round parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Multiple hypothesis corrected sharpened q-values (as in Anderson, 2008) in square brackets. Column 1 (simple means) reports estimates where no additional controls are included. Column 2 includes the lagged baseline covariate as dependent variable. Column 3 presents results from difference-in-differences specifications, and column 4 includes household-level fixed effects in the specification. Sample size as in Table 3.

an overview. Total membership increased during the 2009–11 period, and in the 2011 survey the levels of membership found were substantially higher in the treatment villages than in the control villages.

Table 4 shows that, at the time of the baseline survey in 2009, 6% of the population in both the control and the treatment villages reported that they were members of a VSLA or a similar savings group. Two years later, the figure was 20.7% for the control group and 45% for the treatment group.

This 23.6 percentage point difference in the take-up of the VSLA intervention between the treatment and control villages is significant at the 1% level, suggesting that the randomization was effective in inducing more treatment villagers to participate in VSLAs. However, compliance with the treatment is not perfect, and the randomization suffers from two-sided non-compliance (Gerber and Green, 2012). Not all households living in villages assigned to treatment participated, just as some households from the control villages found their way into VSLA groups. But as Fig. 3 shows, the timing between treatment and control villages differs. Until mid-2009 there was a general, but very small, increase in membership in the area regardless of the random assignment into treatment and control groups. In 2009, when the intervention commenced, membership took off in the treatment villages. In the control villages, however, membership seems to have followed the general pre-project trend until late 2010. This indicates that control group contamination happened relatively late. However, we cannot rule out that the estimated results presented below suffer from some degree of downward bias due to this contamination.

Lønborg and Rasmussen (2014) describe the profile of households who participate in the VSLAs studied here. They suggest that VSLAs do reach very poor households (our study area is one of the poorest areas in one of the poorest countries in the world and so most households in our study area are very poor), but that in the area studied here, the very poorest do not participate.

4.2. Impact on predefined outcomes

The four columns of Table 5 below present the estimated intention-to-treat effects of the VSLA intervention using, respectively, difference-in-means, difference-in-means with lagged value of the dependent variable, pooled difference-in-differences, and first-differenced regression methods on the predefined outcomes. We first discuss the results using the significance levels from individual outcome regressions as the starting point. We then present the interpretation based on the multiple hypothesis q-values.

Overall, we find that the introduction of VSLAs in the treatment villages had a significant impact on a number of outcome variables over the two-year intervention period. Food security improved, particularly in terms of the number of meals consumed the day before the interview. Although the number of income-generating activities decreased, the total volume of savings increased. Finally, there is some evidence of increased total expenditures as predicted using PAT as well as increases in the size of the dwelling, while we do not find any significant impact on the length of the hunger period, on food consumption (measured in terms of the 17 most common food items), on floor quality or a gross asset count.

The increase in total savings arises from an increase in the proportion of people who have liquid savings²⁴, but mostly from an increase in savings amounts among those who already had savings at baseline. The impact on savings is sizeable and corresponds to an increase in savings of 153% in addition to the increase in saving experienced in the control group. Appendix Table A2 shows that there is no evidence

²⁴ In raw numbers 71.4% of households hold liquid savings at baseline, at endline this has increased to 79.3 and 87.5% in control and treatment villages, respectively.

that savings went down in the control group. On the contrary, there is evidence that there was a positive and significant impact on savings among respondents on the list of interested households in control villages, an increase which is entirely driven by VSLA members. There is also no evidence of a decrease in savings among the non-VSLA members. As such, [Appendix Table A2](#) provides no evidence to support worries that control villages households postponed savings while waiting for the VSLA program. Furthermore, when we look into membership into alternative organizations or groups with microfinance components, we find that there is very little crowding out. The introduction of VSLAs in treatment villages reduces membership in village-based groups (some of which may also have a savings and credit function) by 3–4 percentage points.²⁵

The difference-in-means estimates of the impact on the number of income-generating activities are negative across the first two specifications, indicating that households in the treatment villages decreased the number of such activities. *A priori* it is unclear which impact to expect here. The implementing NGO had an aim of increasing the number of income-generating activities of the households involved in the program, but as mentioned in the introduction the fact that a household gains access to savings, credit and insurance against major unforeseen problems may lead to more efficient economic choices, in particular to specialize in a few more profitable income-generating activities rather than to have a diversified set of such activities. A decrease in the number of income-generating activities is consistent with such an explanation, although we cannot identify this to be the exact channel that accounts for the change.

In columns (3) and (4) we control for baseline differences between treatment and control groups by estimating difference-in-differences regressions taking village-level time-invariant characteristics into account as well as household-level time-invariant characteristics using first-differencing. The results across these two specifications are very stable. First of all, the number of meals consumed the day prior to the survey significantly increased: the households from the treatment villages consumed on average 0.13–0.15 more meals per day (s.e. = 0.05 and s.e. = 0.07 respectively), which is equivalent to one in seven households consuming an extra meal. Given that the baseline average is 2.65 meals per day, this is a sizable impact.²⁶

The effect on the number of meals per day is mirrored in a positive effect on overall expenditure as predicted by PAT (marginally significant at the 10% level in the difference-in-differences specification, and at the 5% level for first differences), of about 4.1–4.2% (s.e. = 0.02).

The number of rooms increased by 0.14–0.16 (s.e. = 0.06 and s.e. = 0.07, respectively), which is statistically significant at the 10% level in the difference-in-differences specification, and at the 5% level for first differences. In rural areas in Malawi, where households usually live in huts built of mud plastered onto a wooden frame, it is quite common to build additional rooms onto the existing structure or in the compound when money allows, or – if possible – to replace the mud hut with burnt or un-burnt brick structures.²⁷ However, we found no significant effects on the type of floor in the dwelling. We also found no significant effects on total food consumption measured in terms of the 17 most common food items or on the length of the “hungry” period where households live on less than three meals per day, though the signs of both of these are consistent with an increase in food security.

4.3. Robustness

Our results are weaker in terms of statistical significance and not robust across specifications when we use the multiple hypotheses corrected standard errors. In the simple means specification, the impact on savings and on the number of income generating activities (remains) significant, though at lower levels of significance (5% and 10% levels respectively). In the difference-in-differences specification, not a single variable is significant, while in the first difference specification, our preferred specification for continuous outcomes, the impact on the number of meals consumed yesterday and the size of the house are significant, though only at the 10% level of significance. The impact on expenditure levels as predicted by PAT is not significant at conventional levels of significance, with a q-value of 0.11.

Our conclusion from the multiple hypothesis corrected tests of significance is that there is still evidence of VSLAs having a positive effect on food security, savings and the number of rooms. However, the evidence is much weakened in its statistical significance and in the lack of robustness across specifications.

In [Table 6](#), we assess the robustness of the results to dropping split households, focusing on the estimated intention-to-treat effects. In the 2011 survey, we tracked any households that had moved or split into multiple households during the survey period. Thus, for a number of households, defined from the 2009 survey, we had two observations in the 2011 survey. In the predefined analysis above, we drop one of each of these split households at random from the 2011 sample. In [Table 6](#) we drop all split households, i.e. restricting the sample to intact households. Columns (1) through (4) replicate the specifications of [Table 5](#). The point estimates do not change much, and the significance level of all estimates is nearly identical to the previous specifications.

4.4. Results for intermediate outcomes

In order to investigate possible causal mechanisms for the effects found, we begin by studying whether the intervention actually improved access to and take-up of the financial services offered, namely savings and credit. We then investigate households' agricultural production as well as their business related activities.

4.5. Savings volumes and share-outs

The core component of the VSLA intervention is savings, these being a prerequisite for subsequent credit opportunities and insurance. We showed above the estimated ITT effects on total savings volume. In [Table 7](#) we extend this analysis by looking at the effects on different types of highly liquid savings. We find that the change in overall savings is driven by changes in savings in VSLAs. We do not observe any significant changes in the other savings options (non-VSLA savings as a whole) nor in any of the sub-categories of non-VSLA savings.²⁸

A natural question is where the funds saved in VSLAs come from, since savings in other categories of savings do not decline. The results for consumption suggest that the extra savings do also not come from reduced consumption in terms of meals per day or measured by PAT, our asset-based expenditure estimate.²⁹ The increased VSLA savings may be a result of extra income, just as credit may also play a role. The next sections investigate this further by estimating the impact on credit and on income from business and agriculture.

Before exploring the use of VSLAs, it is important to note that saving in the VSLA groups is not identical to saving in a regular liquid savings account: a core feature of the intervention is the annual share-out of all savings along with any interest earned from loans made during the

²⁵ Results are available upon request from the authors.

²⁶ The impacts on number of meals do not seem to be driven by those households who share-out close to the date of the survey ([Fig. 4](#)). When we drop those households, the impacts on number of meals remain very similar. These results are available upon request from the authors.

²⁷ [Appendix Fig. A1](#) shows that the changes in the number of rooms are primarily due to households in the treatment group adding rooms, as opposed to the housing stock in the control group depreciating.

²⁸ We do not find evidence that VSLAs crowded out ROSCAs. Only 20 households had been a member of a ROSCA at baseline.

²⁹ Note that these consumption measures are unlikely to capture the more hidden consumption used for instance on alcohol or gambling.

Table 6
Regressions without split households.

Outcome	(1) Difference in means no split household	(2) Difference in means with lag and no split household	(3) Difference in difference with no split household	(4) First-differences without split households
Number of months with fewer than three meals a day	−0.204 (0.38)	−0.266 (0.36)	−0.478 (0.45)	−0.541 (0.38)
Number of meals yesterday	0.058 (0.04)	0.071** (0.03)	0.150** (0.07)	0.129** (0.05)
17-Food consumption per week per adult equivalent (MK, log)	0.023 (0.05)	0.043 (0.05)	0.079 (0.07)	0.094 (0.06)
Number of income-generating activities (including agriculture and livestock)	−0.185*** (0.06)	−0.151*** (0.05)	−0.113 (0.11)	−0.068 (0.07)
Total savings (log)	1.043** (0.43)	n.a.	n.a.	n.a.
Per capita expenditure predicted by USAID PAT (log)	0.030 (0.03)	0.035* (0.02)	0.041* (0.02)	0.040* (0.02)
Size of house (number of rooms)	0.102 (0.10)	0.133* (0.07)	0.137* (0.07)	0.159** (0.06)
House has cement floor	0.017 (0.02)	0.001 (0.02)	0.002 (0.02)	−0.010 (0.02)
Asset count	−0.652 (0.63)	−0.475 (0.39)	−0.504 (0.43)	−0.414 (0.36)

Notes: The table presents weighted regressions of predefined outcomes on a treatment dummy, following the same specifications as in Table 5. The sample excludes all households that split between baseline and endline. For total savings, weighted tobit regressions are reported, which is why we do not report difference-in-differences and fixed effects results. Huber–White standard errors clustered at the village level are presented in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 7
ITT effects on savings outcomes.

Outcome	(1) Difference in means
Total savings (log)	1.034** (0.4304)
VSLA savings (log)	3.480*** (0.902)
Non-VSLA savings (log)	0.048 (0.479)
Savings with friend/relative (log)	−0.763 (1.681)
Savings at home (log)	0.192 (0.569)
Savings with bank (log)	0.910 (1.761)

Notes: Table presents results for tobit regressions estimating the impact of VSLA on the log of the value of savings. We implement the suggestion of Cameron and Trivedi (2005) of setting the cutoff at a very small value below the lowest observed value for zero values. Columns 1 presents the results from a tobit regression on the VSLA dummy and block fixed effects. ** $p < 0.05$, *** $p < 0.01$.

cycle. In other words, savings in VSLAs are tied to the VSLA cash box until the date of the share-out, and thus they also become a commitment device. Among the households participating in VSLAs, more than 30% had received a share-out at the time of the survey in 2011.³⁰ Based on households' self-reports, the funds from the share-outs were primarily used for agricultural inputs or investments. This corresponds to the timing of the share-out observed, which was typically during the planting season when there is a need for seeds and fertilizer (see Fig. 4).

³⁰ From the information recorded by the implementing NGO, only 3 of the 102 groups that had been initiated by September 2011 had shared out twice, while another 40 groups had shared out once.

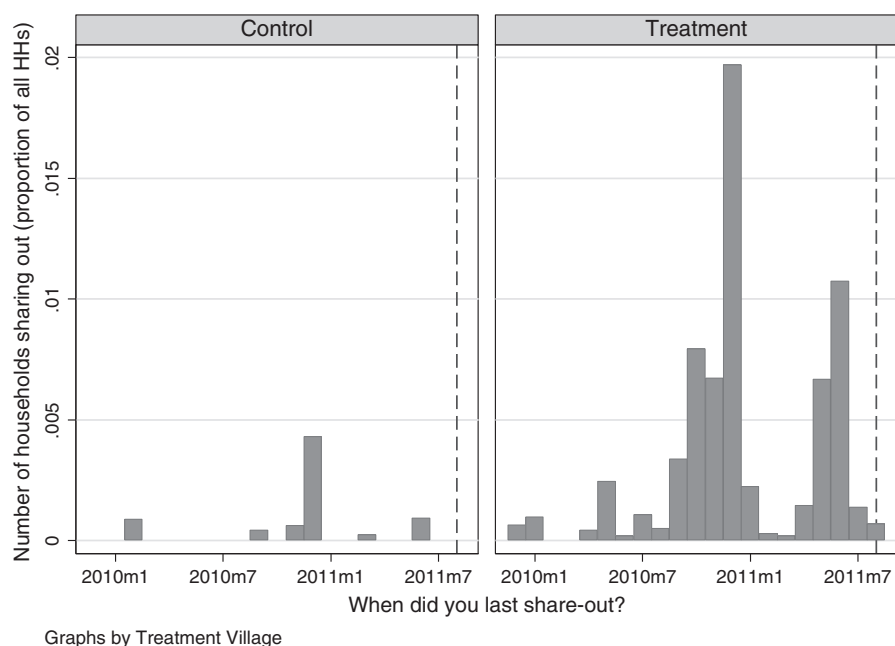
4.6. Credit volume

The other key component of the intervention is the use of pooled savings as credit for the VSLA members.

Table 8 below shows the estimated ITT effects on a range of credit-related outcomes using the same estimation strategies as in Table 5. In addition to increasing the likelihood of taking out at least one loan, living in a treatment village increased the number as well as the value of loans that had been active within the previous twelve months (total loan amount). The intervention also increased the take-up of loans for investment purposes: the probability of having taken a loan for investment purposes and the number of loans taken for investment purposes increased, just as the value of loans taken for investment in agriculture increased significantly. The increases are sizeable and generally significant. Total loan amounts increased significantly and so did the amount borrowed for agriculture; however there was no significant increase in the amount borrowed for business purposes. If we include all types of loan, the raw share of households with loans in treatment villages more than quadrupled from 6% to 26%,³¹ suggesting that access to and use of credit increased substantially.

Although the total loan value for agricultural investments increased considerably more than that for business purposes, more VSLA members report to have taken out loans for trading and other business purposes than for agricultural input or investments. Self-reported use of credit might very well be wrong (Karlan et al., 2015) and thus, we investigate these two potential channels for the effects of VSLAs in the following. Specifically, we look at whether VSLAs seem to have influenced agricultural inputs or outputs and whether VSLAs have influenced other types of business activities.

³¹ Note that this increase is larger than the ITT estimate by itself, as the proportion of control households with loans also doubled, possibly because of spill-over of the intervention into control villages. The proportion of treatment households with any loan at baseline was 5.9% (8.1% for control households). Between baseline and endline, the control villages increased the proportion of households with a loan by 6% and the treatment villages by 19.6%. In total at endline, 25.5% of treatment households had a loan. A little more than 50% of the VSLA members took out loans.



Notes: Dashed vertical line shows median date of endline survey.

Fig. 4. Timing of share-outs (by treatment status).

Table 8
ITT effects on credit outcomes.

Outcome	(1) Baseline mean	(2) Difference in means	(3) Difference in means with lag	(4) Difference-in-difference	(5) First-difference
Household had any loan in past 12 months	0.070 [0.26]	0.122*** (0.04)	0.122*** (0.04)	0.129** (0.06)	0.145*** (0.05)
Number of loans active within past 12 months	0.074 [0.28]	0.138*** (0.04)	0.138*** (0.04)	0.144** (0.06)	0.160*** (0.05)
Total loan amount (log) ^a	8.72 [1.20]	2.714*** (0.76)	n.a. n.a.	n.a. n.a.	n.a. n.a.
Household took out loan for investment purposes in past 12 months	0.044 [0.21]	0.096** (0.04)	0.095** (0.04)	0.076* (0.04)	0.082* (0.04)
Number of investment loans	0.046 [0.22]	0.107*** (0.04)	0.107*** (0.04)	0.086* (0.04)	0.092** (0.04)
Total amount borrowed for agricultural investments (log) ^a	9.38 [0.85]	3.507*** (1.01)	n.a. n.a.	n.a. n.a.	n.a. n.a.
Total amount borrowed for business purposes (log) ^a	8.53 [1.14]	1.183 (0.76)	n.a. n.a.	n.a. n.a.	n.a. n.a.

Notes: Table reports estimates of the impact of VSLA on credit outcomes. Column (1) presents the mean of the outcome at baseline, with standard deviations in square brackets. Column (2) presents the results from the difference in means specification, column (3) from the difference in means with lagged value specification. Columns (4) and (5) present results from difference-in-differences and first differences specifications, respectively. Rows without "a" superscript contain Least squares regressions. Regressions with "a" superscript are tobit regressions, following Cameron and Trivedi (2005) suggestion of replacing zero with the lowest observed value minus a small amount. For these regressions, column 1 presents the estimated mean for observations in the population with non-zero values, whereas the standard deviation represents the sample standard deviation (as some strata do not contain observations at baseline STATA is not able to compute population standard deviations). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.7. Agricultural activities

Since most share-outs are at the end of the year, when purchases of seed and fertilizer are made, Table 9 investigates whether inputs into and outputs of maize have changed. The first column shows the weighted baseline means, whereas columns 2 through 5 provide results for the difference in means, difference-in-means with lag, the pooled difference-in-differences and the first difference regressions, respectively.

On the input side, we find evidence that households in treatment villages were more likely to use fertilizer for growing maize. On the output side, we find a positive effect on the likelihood that households in treatment villages sold part of their maize production and that the total

value of maize sales increased. We do, however, not find evidence that production increased, nor of an increase in maize output. This evidence does point to a change in agricultural practices. However, because this has not led to significantly higher yields, this is unlikely to be the primary channel for the observed increase in food security and expenditures.³²

³² Appendix Table A3 shows the ITT effects on a range of other agricultural inputs and outputs, though we do not find any consistent effect.

Table 9

ITT effects on maize planting practice and outcomes.

Outcome	(1) Baseline mean	(2) Difference in means	(3) Difference in means with lag	(4) Difference-in-difference	(5) First-difference
Household uses any fertilizer on maize	0.452 [.498]	0.135** (0.05)	0.127** (0.05)	0.106* (0.06)	0.093* (0.05)
Area with maize (acres)	1.375 [.897]	−0.306* (0.17)	−0.241 (0.15)	−0.120 (0.15)	−0.154 (0.16)
Quantity of maize harvested (kg, log) ^a	5.759 [.925]	−0.103 (0.13)	<i>n.a.</i> <i>n.a.</i>	<i>n.a.</i> <i>n.a.</i>	<i>n.a.</i> <i>n.a.</i>
Household sold any maize	0.202 [.402]	0.095* (0.05)	0.116** (0.03)	0.140* (0.07)	0.198*** (0.06)
Value of agricultural sale (MK, log) ^a	9.375 [1.55]	−0.582 (0.56)	<i>n.a.</i> <i>n.a.</i>	<i>n.a.</i> <i>n.a.</i>	<i>n.a.</i> <i>n.a.</i>
Value of maize sale (MK, log) ^a	7.936 [.958]	1.489** (0.73)	<i>n.a.</i> <i>n.a.</i>	<i>n.a.</i> <i>n.a.</i>	<i>n.a.</i> <i>n.a.</i>

Notes: Table reports estimates of the impact of VSLA on agricultural inputs. Column (1) presents the mean of the outcome at baseline, with standard deviations in square brackets. Column (2) presents the results from the difference in means specification, column (3) from the difference in means with lagged value specification. Columns (4) and (5) present results from difference-in-differences and first differences specifications, respectively. Rows without “a” superscript contain Least squares regressions. Regressions with “a” superscript are tobit regressions, following Cameron and Trivedi (2005) suggestion of replacing zero with the lowest observed value minus a small amount. For these regressions, column 1 presents the estimated mean for observations in the population with non-zero values. * $p < 0.01$, ** $p < 0.05$, *** $p < 0.01$.

4.8. Business activities

Table 10 reports the estimated ITT effects on income derived from business. This is defined as the money earned (or lost) in the last month of operation, i.e. it is a measure of profit. Based on OLS regressions, the first panel shows that there is some weak evidence that the average number of businesses in the treatment villages increased. Turning to the mean outcome, we see that business income increased by about 6590 MWK (about the size of the share-out) on average (s.e. = 8926). However, due to the very large variation in business income, the estimated standard errors are very large and the result is not statistically significant.

The results in the second panel are therefore based on quantile regressions, giving us results robust to outliers and allowing us to investigate whether there are heterogeneous effects at different place in the business income distribution. We find that median income from business has increased by between 1295 (s.e. = 758) and 1500 MWK (s.e. = 904) more for the median treatment household than for the median control household, an effect that is statistically significant at the 10% level. The treatment effect is higher than the median income from business at baseline. We then investigate whether there are impacts at the 25th and 75th percentiles and find that we cannot reject that there are no effects on average. We then investigate whether business incomes rose for the 66% of households who had already been

Table 10

Effects on small business outcomes.

Outcome	Baseline mean	Difference in means	Difference in means with lag
<i>OLS regressions</i>			
Number of businesses (excluding agriculture and livestock)	0.93 [0.778]	0.192 (0.129)	0.241* (0.125)
Total income from all businesses (MWK)	10523.2 [49834.1]	6589.8 (8926.315)	2453.4 (5609.115)
<i>Quantile regressions</i>			
Business income all respondents	25th	0 (463.3)	386.5 (435.8)
	50th (median)	700 (904.4)	1295* (758.3)
	75th	4000 (1747)	1652 (1559)
	25th	723 (730.2)	615.0 (694.2)
	50th (median)	2500 (1330)	2811** (1208)
	75th	4600* (2407)	3911* (2069)
Business income respondents with business at baseline	25th	0 (123.1)	<i>n.a.</i>
	50th (median)	0 (1108)	<i>n.a.</i>
	75th	−2400* (1421)	<i>n.a.</i>
Business income respondents without business at baseline	75th	0 (1421)	<i>n.a.</i>

Notes: Table reports estimates of the impact of VSLA on income generating activities. Column 1 presents the mean of the outcome at baseline, with standard deviations in square brackets. Columns 3 and 4 present results from difference in means and difference in means with lag, respectively. OLS regressions follow specifications described in Table 5. Quantile regressions include stratification dummies, as well as a dummy for baseline business activity. Quantile standard errors are cluster-bootstrapped and include the randomization strata. Sample: long questionnaire respondents. * $p < 0.1$, ** $p < 0.05$.

involved in a small-scale business at baseline. Here we find more precisely estimated treatment effects, suggesting significant positive impacts on both the 50th and 75th percentiles. Conversely, for households who had not yet been engaged in some small-scale business, there is no evidence that their business income increased for the median household.

Overall, the evidence suggests that the program seems to have led to significant increases in investments in agriculture, although we cannot detect any impact on agricultural outcomes. On the other hand, the VSLAs were more successful in increasing the income from existing small-scale businesses.

4.9. Cost effectiveness

In order to assess whether VSLAs are worthwhile, this section provides some simple cost-effectiveness calculations. We estimate the cost per year of the implementation in the project studied as total costs (USD 201,000) divided by three. The specific intervention investigated was quite costly compared to the average VSLA intervention due to its relatively small scale and the fact that VSLAs often use village agents to a greater extent than the randomization allowed in the project setup.³³

The implementing partner reported having a total number of 1783 VSLA members by September 2011 out of the 3800 households in the treatment villages. The cost per member was thus USD 75,³⁴ while the cost per household in a treatment village was USD 35 per year of implementation.

The estimated effect on total household expenditures of 4.2 per cent corresponds to an increase in the average household expenditures of 0.124 USD per household per day, which is equivalent to USD 0.31 at PPP adjusted prices (PPP 2005). On this basis the project broke even after 280 days or a bit over nine months of impact (based on the two-year impact estimates).³⁵ These estimates do not include impacts beyond the duration of the two years or impacts on other measures beyond the impact on household expenditures.

5. Conclusion

This paper provides evidence that improvements in local financial market intermediation can have a significant impact on household welfare and economic activities in remote rural areas of developing countries. Village Savings and Loan Associations have been heavily promoted in the hope of remedying what has been termed the “last mile problem of microfinance”: operating a microfinance institution and (even more so a bank) is rarely profitable in remote rural areas in developing countries. VSLAs, on the other hand, do not rely on any outside injection of funds or professional management once set up. They harness savings in the local community which are re-invested locally, and are entirely self-managed. A set of carefully thought-out safeguards and governance features seem to be able to sustain the trust necessary for an improved local financial market intermediation. As household wealth increases and VSLAs mature, it is possible that these semi-formal institutions might even be integrated into the formal financial sector.

This paper provides evidence of the effects on household welfare of introducing VSLAs in rural Malawi. We find that this introduction raises welfare along a number of dimensions, in particular for improved food

security and strengthened household income indicators. The results provide evidence that households in the poorest, most rural areas are constrained by lack of access to financial institutions providing savings and credit opportunities. Having analyzed the impact only two years after the onset of the project, we find these indications of the positive effects of simply organizing savings groups which also create credit opportunities for their members rather encouraging.

In studying how the money from share-outs and credit is used, we find that the use of fertilizer rises, and that loans are used primarily in agriculture and small-scale businesses. Agricultural output increases in response to the increased investments; however we cannot detect any impact on agricultural outcomes. On the other hand, we do see an increase in total business income among households, which already had a business at baseline, whereas there was no significant increase in business income among households with newly established businesses. Judging by these results, the provision of opportunities for the village to save and take out loans for agricultural and business investments may play a key role in travelling “the last mile”.³⁶ The mechanism through which VSLAs might have impacts in different contexts might well differ, given the different roles participants can play in these savings groups. The mechanism we identify suggests the presence of financial market imperfections, whereas Beaman et al. (2014) suggest that women in their treatment villages become more forward looking.

The above conclusions have limitations, however, which should be kept in mind. Out of the 80,000 VSLAs across the world, we have studied only a few. The results may be specific to the context, cultural and economic, in which they appear, and the particular constraints faced by households in these communities. Future assessments from other places, including the ones we know are under way, will shed light on the extent to which these findings can be generalized.

Appendix A. Protocol of intervention of NGO in the villages

Both the villages in the treatment and control group received the same information, according to the field reports from the implementing agency. Two types of meetings were held, following the VSLA manual from 2009.

1. One orientation of community leaders and administration officials.
2. Introducing VSL to the community.

The first meeting included Senior Chiefs, Group Village Headmen, Village Headmen, religious leaders, various development committees currently present in the project area as well as government workers. The objective of this meeting was to obtain the support of key stakeholders. The agenda was the following:

1. Introduction of the implementing organisation.
2. Project goals and objectives.
3. Target group to be served.
4. Services offered.
5. Role of local leaders and administrators.
6. The stakeholders were told about similar programs in nearby regions (that had been successful).

The second set of meetings was carried out in each of the 46 villages. The goal of these meetings was to create general awareness in the community of the programme's purpose, methodology and process and to offer the opportunity to register for training. The purposes were described to be to enable villagers to manage their own savings and to administrate a loan facility through which they could earn a profit. It was emphasized that there would be no injection of funds into the saving groups other

³³ Normally, VSLAs are introduced to a single village. Some VSLA members then become village agents, who subsequently start VSLAs in neighboring villages. While this limits the costs of the intervention, we do not know what the effects of relying on village agents rather than trained field officers would be.

³⁴ This is higher than the typical cost of implementing VSLAs as reported by Allen and Panetta (2010), which was USD 18 to 48 per member.

³⁵ The estimates are very close, when instead we use the treatment effect on meals per day of 0.126 meals per day as a starting point.

³⁶ We do not know to what extent participation and savings in VSLAs are driven by the commitment aspect of the associations—the obligation to contribute a minimum amount each week. Duflo et al. (2011) suggest that in the Kenyan context, the inability to save for investments is due to behavioral biases.

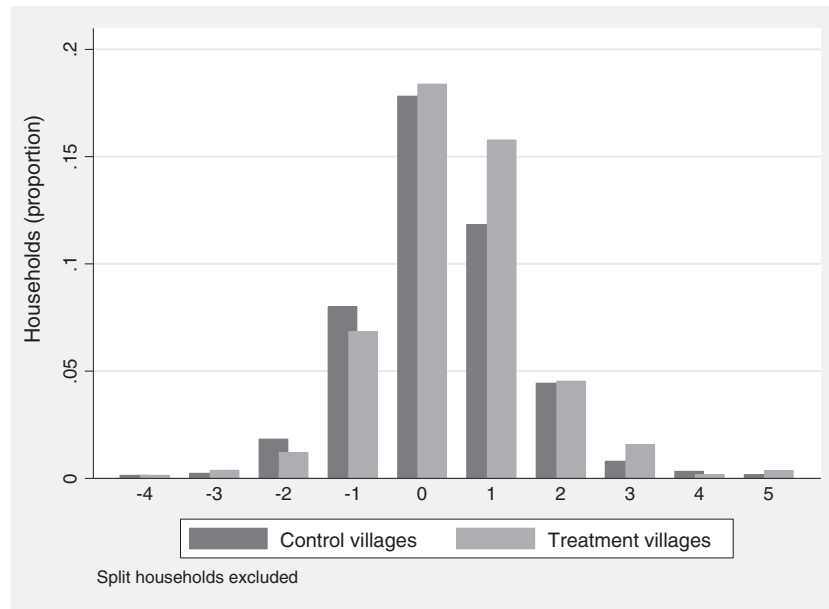


Fig. A1. Change in number of rooms over time

than a cash kit, which would be issued out as a loan. All villages were informed that 50 groups would be formed each year for three years and not all villages could receive training in the first year of implementation. Rather a lottery draw would be conducted to find which villages would be included in the project between 2009 and 2011 and which would be included after 2011. It was in these meetings that villagers could voice their interest and put their names on the initial list of interested households. Following meetings in all 46 villages, the randomization was carried out and an introductory meeting was held with groups in the 23 treatment villages.

Appendix B. Multiple hypothesis testing procedure

The usual p-values are underestimated when multiple hypotheses are tested, and need to be adjusted so that the rejection rate is correct under the null. To gain an intuition of how this works, take the case where you are testing two variables, X and Y with p-values of 0.045 and 0.04. The standard Bonferroni correction would say that since you are testing two variables, you have (almost) twice the probability of finding at least one error at the 5% level of significance, so you have to “adjust” the p-values by a factor of 2. In this example, such an adjustment would mean that the variables are not significant at the 5% level of significance, even though *both* are individually significant at that level.

An alternative adjustment procedure is suggested by Benjamini and Hochberg (1995). Their method proceeds as follows: it orders the p-values from the lowest to highest. Define p_i as the i th p-value in this ordering. Then analyse whether $p_i < i/m * \alpha$, where i is the ordering parameter, m are the number of total hypotheses tested, and α the level of significance. If the inequality is met for any variable, then all variables with a lower p-value are considered significant for significance level α . In the example of X and Y the inequality is not met for the first variable, but is true for the second variable, so both variables are significant at the 5% level.

Benjamini et al. (2006), as explained in Anderson (2008), show how to improve power further. They implement a two-stage procedure, where the idea is as follows: the cut-off from the Benjamini and Hochberg method mentioned above (whether $p_i < i/m * \alpha$) assumes that all null hypotheses are true. If say c of the first variables is

significant at very, very high levels of significance, we should take that into account. This can be done as follows (see Anderson, 2008): Implement the Benjamini and Hochberg method with a significance level of $\alpha/(1 + \alpha)$. Define c as the number of rejected hypotheses. Compute $m_0 = m - c$, which is an estimate of the number of incorrect hypotheses. Redo the Benjamini and Hochberg method with the following cutoff: $p_i < i/m_0 * \alpha/(1 + \alpha)$. Anderson (2008) then shows how to go from rejection at specified levels of significance to the lowest level of significance, at which the hypotheses can be rejected, these adjusted p-values are denoted as sharpened q-values.

Table A1
VSLA interest and participation.

Dep var: VSLA membership	(1) Linear Probability Model
Panel 1 HHs not on initial list of interested in control villages	0.113
Panel 2: Increase in participation over base category (HHs not on initial list of interested in control villages)	
On interested list in control villages	0.152*** (0.0334)
Not on interested list in treatment villages	0.241*** (0.0398)
On interested list in treatment villages	0.491*** (0.0439)
Panel 3: Interaction between Interest and treatment (coefficient and significance)	
On interested list * treatment village	0.098** (0.0455)
Observations	1680
R-squared	0.107

Notes: Table reports estimates of probability of participation in VSLA using a Linear Probability Model. Panel 1 presents the proportion of households who participate in VSLAs. This is the base category. Panel 2 presents estimates of the increase in the participation probability for the three categories: On interested list in control villages, not on interested list in treatment villages, and on interested list in treatment villages. The estimated coefficients are also from a Linear Probability Model that includes block fixed effects. Panel 3 presents the estimate of whether the effect of being both interested and in a treatment village (0.491) at the same time is different from the sum of the coefficients of being interested (0.152) and being treated (0.241). Robust standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

Table A2

Time trend in savings in control villages.

	(1)	(2)	(3)	(4)	(5)
Dep var: Log(savings + 1)	Entire control group	Control villages: on list of interested HH	Control villages: not on list of interested HH	Control group VSLA members	Control group non-VSLA members
Endline	0.795 (0.798)	1.317*** (0.409)	0.687 (0.945)	2.906*** (0.577)	-0.038 (0.942)
Observations	811	509	302	267	544
R-squared	0.018	0.045	0.018	0.234	0.016

Notes: Table reports estimates of a regression of savings on a dummy for endline, as well as block fixed effects. Column 1 presents results for the entire sample of control group members. Column 2 shows the estimate from a regression of savings on a dummy for endline for HHs on the interested list. Column 3 present equivalent results for HHs not on the interested list. Column 4 presents results for respondents who in either of the two survey rounds were members of a VSLA. Column 5 presents results for respondents who were never VSLA members. The dependent variable is different than in Table 5, as implementing a tobit regression in the specification of Table 5 results in an error, due to one strata only containing a single village in the control group. Robust standard errors in parentheses. ***p < 0.01, **p < 0.05, *p < 0.1.

Table A3

Effects on other agricultural outputs.

	(1)	(3)	(2)	(4)	(5)
Outcome	Baseline mean	Difference in means	Difference in means with lag	Difference-in- difference	First-difference
Household uses any fertilizer	0.702 [0.458]	0.088 (0.06)	0.092 (0.05)	0.108 (0.08)	0.1 (0.07)
Household cultivates vegetable plot	0.247 [0.431]	-0.047 (0.03)	-0.023 (0.03)	0.037 (0.04)	0.037 (0.03)
Household has any irrigated plots	0.28 [0.449]	0.01 (0.05)	0.047 (0.05)	0.125 (0.08)	0.166** (0.07)
Household used any purchased seeds	0.803 [0.398]	0.011 (0.05)	0.017 (0.05)	0.054 (0.07)	0.073 (0.08)
Total area cultivated (acres)	2.876 [1.716]	-0.251 (0.19)	-0.119 (0.16)	-0.013 (0.21)	0.01 (0.21)
Area with local maize (acres)	0.53 [0.784]	-0.309** (0.15)	-0.286* (0.15)	-0.183 (0.12)	-0.227* (0.13)
Area with composite maize (acres)	0.079 [0.36]	-0.009 (0.01)	-0.009 (0.01)	-0.022 (0.03)	-0.02 (0.02)
Area with hybrid maize (acres)	0.766 [0.86]	0.012 (0.08)	0.034 (0.08)	0.084 (0.08)	0.094 (0.07)
Area with tobacco (acres)	0.184 [0.439]	0.023 (0.06)	0.043 (0.05)	0.06 (0.06)	0.071 (0.06)
Area with cotton (acres)	0.315 [0.64]	-0.022 (0.06)	-0.021 (0.04)	0.002 (0.07)	-0.022 (0.06)
Area with rice (acres)	0.316 [0.593]	-0.18 (0.13)	-0.098 (0.09)	-0.038 (0.11)	-0.015 (0.11)
Local maize harvested (kg, log) ^a	5.397 [0.913]	-0.920* (0.53)	n.a.	n.a.	n.a.
Composite maize harvested (kg, log) ^a	5.911 [1.008]	0.292 (0.61)	n.a.	n.a.	n.a.
Hybrid maize harvested (kg, log) ^a	5.668 [0.957]	0.33 (0.28)	n.a.	n.a.	n.a.
Local maize harvested per acre (kg, log) ^a	5.35 [0.8]	-0.816 (0.50)	n.a.	n.a.	n.a.
Composite maize harvested per acre (kg, log) ^a	5.741 [0.897]	0.417 (0.72)	n.a.	n.a.	n.a.
Hybrid maize harvested per acre (kg, log) ^a	5.639 [0.837]	0.374 (0.28)	n.a.	n.a.	n.a.
Household sold any crops	0.592 [0.492]	-0.041 (0.06)	-0.021 (0.06)	-0.016 (0.08)	0.027 (0.07)

Appendix C. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jdeveco.2015.12.003>.

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Notes: Table reports estimates of the impact of VSLA on agricultural inputs. Column (1) presents the mean of the outcome at baseline, with standard deviations in square brackets. Column (2) presents the results from the difference in means specification, column (3) from the difference in means with lagged value specification. Columns (4) and (5) present results from difference-in-differences and first difference specifications, respectively. Rows without “a” superscript contain Least squares regressions. Regressions with “a” superscript are tobit regressions, following Cameron and Trivedi (2005) suggestion of replacing zero with the lowest observed value minus a small amount. For these regressions, column 1 presents the estimated mean for observations in the population with non-zero values. *p < 0.10, **p < 0.05, ***p < 0.01.